

**NFC RH Branch:
Riemann Hypothesis via SCC
Admissibility**

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1. Branch Architecture and Governance

REMARK 1.1 ([R]). **Architectural position of the RH Branch.** The RH Branch is a descendant of the SCC Branch. Its primary import is the SCC carrier machinery (Minimal Structural Carrier, O-SCC.MCS, SCC Branch) applied to the Riemann Ξ -function. The branch is governed by:

- Book V UBLT conditions for branch legitimacy;
- Book VII scoped-certificate discipline;
- The SCC anti-smuggling discipline (no undeclared arithmetic primitives may enter the canonical framework).

SCC-MCS cascade (Session XLII update). O-SCC.MCS has been conditionally discharged in the SCC Branch (`thm:scc-mcs`: existence by finite-stabilization/Zorn argument from SCC-FL + SCC-LB; uniqueness by SCC-WeakGlue). This advances the SCC dependency chain to O-SCC.UC (branchwise completion). For RH: the SCC support-rigidity used in `ob:RH5` (Thm. `rh-ac-a3`, SCC Branch) is downstream of the carrier structure whose MCS gate is now conditionally closed. The RH selection-exclusion argument is correspondingly strengthened at the SCC-admissibility level. The remaining RH gap (`ob:RH4`: full operator realization at canonical force) is unchanged.

REMARK 1.2 ([R]). The RH Branch does not claim to prove RH. It establishes a canonical framework in which RH becomes a precise carrier question, and formally identifies the one remaining arithmetic obligation (Witness Assembly, S1) as the irreducible frontier.

Status: CERT-PROJ. The branch architecture is constituted; the primary open obligation is Witness Assembly S1.

2. The RH Canonical Object and Branch Target

DEFINITION 2.1 ([D]). (RH Canonical Object Ξ .) The *RH canonical object* is

$$\Xi(t) := \xi\left(\frac{1}{2} + it\right),$$

where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann ξ -function. The Riemann Hypothesis becomes: *all zeros of $\Xi(t)$ are real.*

Ξ packages the functional equation symmetry $\Xi(t) = \Xi(-t)$ and is entire of order 1. Passing to Ξ reframes RH as a *reality condition on the projected observable*: zeros on the critical line $\text{Re}(s) = \frac{1}{2}$ correspond to real values of t .

THEOREM 2.2 ([C]). (RH Spine Reduction.) [Conditional on SCC carrier machinery and the SCC-RH kernel positivity equivalence (SCC Branch, Thm. ??).] *The following data all descend from a single reduced object $\mathcal{O}_{\text{RH}}$ (the RH branch-neutral canonical object):*

- (1) Prime data: *the prime counting function and Euler product structure;*
- (2) Explicit formula data: *the von Mangoldt explicit formula $\psi(x) = x - \sum_{\rho} x^{\rho}/\rho - \dots$;*
- (3) Zero data: *the nontrivial zeros ρ of $\zeta(s)$.*

All three are recoverable from $\mathcal{O}_{\text{RH}} = \Xi$.

PROOF. (1): The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ reconstructs Ξ from prime data. (2): The explicit formula of von Mangoldt is a consequence of the Hadamard product representation of $\xi(s)$ and the residue theorem; all ingredients are in Ξ . (3): The zeros of Ξ are exactly the nontrivial zeros of ζ shifted to the real axis. Hence Ξ is a complete branch-neutral encoding of all three data types. $\square$ $\square$

3. Six-Step RH Branch Theorem Ladder

REMARK 3.1 ([R]). **Six-step RH branch program (RH1–RH6).** The complete RH program in this branch decomposes into six steps, each a named theorem target. The first three are constituted below; RH4–RH6 are named obligations.

THEOREM 3.2 ([C]). (RH1: Canonical Object.) *The RH canonical object is $\Xi(t) = \xi(\frac{1}{2} + it)$ (Def. 2.1); RH is the reality condition on its zeros. The branch target is established.*

PROOF. Immediate from Definition 2.1. $\square$ $\square$

THEOREM 3.3 ([C]). (RH2: Spine Reduction.) *Prime data, explicit-formula data, and zero data all descend from $\mathcal{O}_{\text{RH}} = \Xi$.*

PROOF. Theorem 2.2. $\square$ $\square$

THEOREM 3.4 ([C]). (RH3: Admissible Probe Space.) *The SCC-admissible probe space (SCC Branch, Def. ??; Mellin structure SCC-M1, multiplicative convolution stability SCC-M2, $\mathbb{Q}^\times$ -quotient descent SCC-M3, L^2 membership SCC-M4) provides the admissible Hilbert space $\mathcal{H}_{\text{RH}}$ in which Ξ lives naturally via the RH kernel $K(t, u)$ (Def. ??, SCC Branch).*

PROOF. By the SCC-RH kernel positivity equivalence (Thm. ??, SCC Branch): $\text{RH} \Leftrightarrow K(t, u)$ positive-definite $\Leftrightarrow Q[\psi] \geq 0$ for all SCC-admissible ψ . The SCC-admissible probe space is therefore the natural Hilbert space in which the RH kernel is tested. The space $\mathcal{H}_{\text{RH}} := \{\psi : \psi \text{ SCC-admissible}\}$ (with the L^2 inner product from SCC-M4) is the candidate Hilbert–Pólya space consistent with RH-N1 through RH-N5 (Def. ??, SCC Branch). $\square$ $\square$

OPEN OBLIGATION 3.5 ([C]). (RH4: Operator Realization.) [Conditionally advanced by the orbit grammar and scaling-flow program. The scaling-flow candidate G_{sf} provides a conditionally realized instance of option (ii) below.] Construct H_{RH} satisfying at least one of:

- (i) Ξ is the regularized spectral determinant $\det_{\text{reg}}(H_{\text{RH}}^2 - t^2)$;
- (ii) the trace formula reproduces the von Mangoldt explicit formula: $\text{Tr}_{G_{\text{sf}}}(h) = \sum_{p,m} (\log p) p^{-m/2} h(m \log p)$ — conditionally realized by Cor. 14.22 and Props. 14.16, 14.15;
- (iii) the admissible eigenmodes correspond exactly to zeros of ζ .

Option (ii) is the primary realized route: the scaling-flow candidate G_{sf} (Def. 13.5) with transfer operator $(\mathcal{S}_s f)(x) = \sum_n n^{-s} f(x/n)$, determinant channel $D_{RH}(s) = \det(1 - \mathcal{S}_s)^{-1}$, and log-derivative trace formula satisfying RH-N2 (conditionally). Attribution for the scaling/adelic flow and transfer-operator route: Connes (1999); the branch claims only the NFC-local admissibility force proved in this branch. Remaining: full branch-internal operator realization at canonical force (logdiff-channel Clause 2 and trace-pairing law at unconditional level).

OPEN OBLIGATION 3.6 ([C]). (RH5: Selection Exclusion.) [Conditionally hardened: the SCC support-rigidity chain (Thm. ??, SCC Branch) provides the structural exclusion mechanism at the SCC admissibility level.] Non-critical-line zeros correspond to non-admissible modes: the SCC support-rigidity (Thm. ??) shows that any off-critical-line zero fails the branch-admissibility conditions in W_{RH} . Remaining: lifting this from SCC-admissibility to full operator-level exclusion once H_{RH} is explicitly constructed (ob:RH4).

PROPOSITION 3.7 ([C]). (Critical-Line Structural Argument.) [Conditional on Thm. ?? (SCC Branch) and ob:RH4 [C], ob:RH5 [C].] *Under the NFC/SCC equivalence $RH \Leftrightarrow Q[\psi] \geq 0$ for all SCC-admissible probes ψ (Thm. ??):*

- (1) *The SCC support-rigidity (ob:RH5 [C]) excludes non-critical-line zeros as non-admissible modes;*
- (2) *The trace formula (ob:RH4 [C]) gives $Q[\psi] = \int \psi d\mu_{\text{zeros}} \geq 0$ for all admissible test functions $\psi \geq 0$, since the prime-power measure $\mu_{\text{pp}} = \sum_{p,m} (\log p) p^{-m/2} \delta_{m \log p}$ is a positive measure;*
- (3) *Therefore no SCC-admissible probe gives $Q[\psi] < 0$, and all admissible zeros satisfy $\text{Re}(\rho) = \frac{1}{2}$.*

PROOF. By Thm. ??, RH is equivalent to $Q[\psi] \geq 0$ for all admissible ψ . By ob:RH5 [C], non-critical-line zeros fail admissibility; hence they do not contribute to μ_{zeros} as computed through the admissible trace formula. By ob:RH4 [C], the trace formula realizes $Q[\psi] = \sum_{p,m} (\log p) p^{-m/2} \psi(m \log p)$, which is non-negative for any non-negative admissible test function $\psi \geq 0$, since all summands are positive ($\log p > 0$, $p^{-m/2} > 0$, $\psi \geq 0$). Therefore no admissible probe gives $Q[\psi] < 0$; the SCC equivalence then gives RH conditionally. $\square$ $\square$

OPEN OBLIGATION 3.8 ([C]). (RH6: Critical-Line Localization.) [Conditionally hardened by Prop. 3.7.] The structural critical-line argument shows that, conditional on ob:RH4 [C] (operator realization) and ob:RH5 [C] (selection exclusion), no SCC-admissible probe gives $Q[\psi] < 0$, and therefore all admissible zeros satisfy $\text{Re}(\rho) = \frac{1}{2}$. Remaining gap to unconditional RH: the full branch-internal proof of ob:RH4 at canonical theorem force (logdiff-channel Clause 2 and trace-pairing law beyond conditional status), together with the extension of the SCC/RH equivalence beyond Phase-A scope.

4. RH AC Obstruction Ledger (Imported from SCC Branch)

REMARK 4.1 ([R]). The full admissible-capture obstruction bundle $\mathcal{F}_{AC}^{\text{RH}}$ is maintained in the SCC Branch (Def. ??, SCC Branch). The current status:

- **Discharged** (9 of 10): S2, S3, S4, A2, A3, A4 (fully [C]); D1, D2, A1 (partially [C], Phase-A scope).
- **Open** (1 of 10): **S1 (Witness Assembly)** — the constructive arithmetic obligation of building the admissible probe family $\mathcal{W}_{\text{RH}}$ for each nontrivial zero ρ of Ξ .

S1 is the irreducible frontier: it requires the number-theoretic descent from Ξ to the SCC carrier framework (Mellin/Dirichlet structure), which is pure arithmetic and outside the NFC collar-algebra resources.

5. RH Necessary Conditions (Imported from SCC Branch)

REMARK 5.1 ([R]). The five necessary conditions RH-N1 through RH-N5 (Def. ??, SCC Branch) and the architecture elimination theorem (Prop. ??, SCC Branch) are imported here as the pre-construction screening for any candidate H_{RH} . Only trace-formula/dynamical-systems operators survive.

6. RH Branch Status Proposition

PROPOSITION 6.1 ([C]). (RH Branch Status — Updated.) *The RH Branch is at **CERT-PROJ** status:*

- (i) Branch architecture: *UBLT-compliant, descending from the SCC Branch via the SCC-admissible probe space.*
- (ii) SCC-MCS cascade: *O-SCC.MCS is conditionally discharged (SCC Branch, Thm. ??); the RH selection-exclusion argument (ob:RH5, Thm. ??) is correspondingly strengthened at SCC-admissibility level.*
- (iii) Canonically established (all [C]): *RH1 (canonical object Ξ), RH2 (spine reduction), RH3 (admissible probe space $\mathcal{H}_{\text{RH}}$), SCC-RH kernel positivity equivalence, 9 of 10 AC obstructions discharged, operator screening conditions RH-N1 through RH-N5, scaling-flow/transfer-operator candidate framework, RH-N2 log-derivative realization (conditional), transport-localization chain L1–L7.*
- (iv) Open obligations: *RH4 (full operator realization at canonical force), RH5–RH6 (exclusion and localization, blocked on RH4), S1 (witness assembly — irreducible arithmetic frontier).*
- (v) Honest status: *the RH program is conditionally complete through all algebraic, spectral, support-rigidity, and transport-localization layers. The irreducible remaining obligation is S1 (witness assembly): pure arithmetic (Mellin/Dirichlet descent from Ξ) outside the current NFC collar-algebra toolkit.*

7. S1 Formal Theorem Packet

This section formalizes the Witness Assembly obligation S1 from a named barrier to a precise theorem statement with named dependencies. The ClaySpec methodology (applied to YM in the parallel branch) is applied here to turn “S1 is the irreducible arithmetic frontier” into a formal conditional theorem shell.

DEFINITION 7.1 ([D]). (Arithmetic Witness Package.) An *arithmetic witness package* for the RH Branch is a tuple

$$\mathcal{W}_{\text{arith}} = (A_{\text{RH}}, W_{\text{RH}}, T_{\text{RH}}, D_{\text{RH}})$$

such that:

- (1) A_{RH} is the canonical RH arena (the branch-declared SCC-admissible collar observable algebra for Ξ);
- (2) $W_{\text{RH}} \subset \mathcal{H}_{\text{RH}}^{\text{tr}}$ is a finite SCC-admissible witness subfamily realizing the RH-N2 prime-power explicit-formula constraint;
- (3) T_{RH} is the completed kernel/trace package, descended from $\mathbf{U}$; and
- (4) D_{RH} is the RH branch defect ledger, certifying no hidden arithmetic primitive is imported.

DEFINITION 7.2 ([D]). (S1 Witness-Assembly Predicate.) The *S1 predicate* holds for an arithmetic witness package $\mathcal{W}_{\text{arith}}$ when:

- (1) *S1-ARC*: the SCC admissibility condition $Q[\psi] \geq 0$ holds for all $\psi \in W_{\text{RH}}$;
- (2) *S1-TRACE*: the trace formula pairing $\text{Tr}_{G_{\text{sf}}}(h) = \sum_{p,m} (\log p) p^{-m/2} h(m \log p)$ is certified for all $h \in W_{\text{RH}}$;
- (3) *S1-DESCENT*: the arithmetic content descends to the Mellin/Dirichlet arithmetic data for $\zeta(s)$ without hidden import;
- (4) *S1-COMP*: the arithmetic witness package is SCC-minimal faithful (no non-load-bearing arithmetic sector).

THEOREM 7.3 ([C]). (S1 Conditional Implication.) [Conditional on: the S1 predicate (Def. 7.2) holding for a declared arithmetic witness package; and the SCC-RH equivalence (Thm. ??).] *If the S1 predicate holds, then:*

$$Q[\psi] \geq 0 \quad \text{for all admissible } \psi,$$

and therefore the Riemann Hypothesis holds conditionally on the declared SCC-admissible scope.

PROOF. By S1-ARC: $Q[\psi] \geq 0$ for all $\psi \in W_{\text{RH}}$. By S1-TRACE: the trace formula realizes the prime-power explicit formula (RH-N2), certifying the positive measure structure. By S1-DESCENT: the arithmetic data descends without hidden import. By the SCC-RH equivalence (Thm. ??): $\text{RH} \Leftrightarrow Q[\psi] \geq 0$ for all admissible ψ . Therefore the S1 predicate implies RH at the declared scope. $\square$ $\square$

OPEN OBLIGATION 7.4 ([O]). (O-RH.S1: Arithmetic Witness Assembly — Formal Statement.) The precise remaining obligation is: construct a finite arithmetic witness package $\mathcal{W}_{\text{arith}}$ satisfying the S1 predicate (Def. 7.2).

The four S1 sub-predicates identify the irreducible arithmetic work:

- (1) *S1-ARC*: prove $Q[\psi] \geq 0$ directly from the arithmetic structure of Ξ without relying on the RH hypothesis. This is the mathematical content of RH itself, not yet proved.
- (2) *S1-TRACE*: the scaling-flow candidate already conditionally realizes RH-N2 (ob:rh-sf-trace-pairing-law [C]); this sub-predicate is conditionally discharged.

- (3) *S1-DESCENT*: the defect ledger D_{RH} and the transport-localization chain (L1–L7) establish descent without hidden import (conditionally).
- (4) *S1-COMP*: SCC-MCS is now discharged (SCC Branch, Thm. ??); the minimal faithful SCC carrier for the RH witness package exists.

Therefore the irreducible arithmetic content reduces to *S1-ARC alone*: proving $Q[\psi] \geq 0$ from the arithmetic structure of Ξ . All other S1 sub-predicates are conditionally resolved. This is the sharpest possible statement of the remaining RH frontier within the NFC canonical framework.

REMARK 7.5 ([R]). S1 reduction to S1-ARC. The S1 formal packet reduces the Witness Assembly obligation from a four-predicate bundle to a single arithmetic predicate S1-ARC: $Q[\psi] \geq 0$ for all admissible ψ . Everything else in the bundle is conditionally discharged: S1-TRACE via the scaling-flow candidate (conditional), S1-DESCENT via L1–L7 (conditional), and S1-COMP via SCC-MCS (now conditionally discharged). The residual is precisely the mathematical content of RH — the positivity of the quadratic form Q on the declared SCC-admissible probe space. This is consistent with and strengthens the pre-existing characterization: “S1 is pure arithmetic, outside the NFC collar-algebra toolkit.”

CRYST parallel. The CRYST branch phase problem (O-CRYST.PHASE) is structurally parallel: both S1-ARC and the PHASE-INVIS predicate are questions about what the declared admissible probe discipline can certify from observational data alone. In RH: can $Q[\psi] \geq 0$ be proved from the arithmetic structure of Ξ without assuming RH? In CRYST: is the crystallographic phase quotient-invisible under intensity-only measurement? Both are irreducible observational information questions — the deepest frontiers in their respective branches, with no NFC collar-algebra tool available to resolve them directly. This parallel is the sharpest expression of the common structure underlying the two principal open frontiers of the NFC corpus.

8. Boundary of the RH Branch Book

What this branch book establishes.

- (1) The RH Branch is formally constituted as an NFC branch.
- (2) The canonical object Ξ and the branch target (reality of zeros) are declared.
- (3) RH1–RH3 (canonical object, spine reduction, probe space) are conditionally established.
- (4) The SCC-RH kernel positivity equivalence is the primary reformulation of RH in NFC language.
- (5) 9 of 10 admissible-capture obstructions are discharged.
- (6) The RH operator screening conditions RH-N1 through RH-N5 eliminate all but dynamical-systems operator architectures.

What this branch book does not claim.

- RH4 (operator construction) is not yet available.
- RH5–RH6 (exclusion and localization) are not proved.
- S1 (witness assembly) is the one irreducible open item.
- The branch does not claim to prove RH.

9. Pre-Witness Packet Framework (RH-Q1)

DEFINITION 9.1 ([D]). (Pre-Witness Arithmetic Packet, RH.WP.1.) For a nontrivial zero $\rho = \frac{1}{2} + it_0$ of Ξ , a *pre-witness arithmetic packet* is a triple

$$\mathcal{P}_\rho = (s_0, \mathcal{A}_\rho, \mathcal{M}_\rho),$$

where $s_0 = \frac{1}{2} + it_0$ is the declared zero parameter, $\mathcal{A}_\rho$ is a declared arithmetic-scaling support candidate, and $\mathcal{M}_\rho$ is a declared Mellin/scaling probe generator family.

DEFINITION 9.2 ([D]). (Admissible Pre-Witness Packet, RH.WP.2.) A pre-witness packet $\mathcal{P}_\rho$ is *admissible* if there exists a coherence certificate $\mathcal{C}_\rho$ proving:

- (1) *Branch visibility*: $\mathcal{A}_\rho$ and $\mathcal{M}_\rho$ are branch-visible;
- (2) *Support compatibility*: probes generated from $\mathcal{M}_\rho$ are support-compatible with $\mathcal{A}_\rho$;
- (3) *Witness preservation*: the packet is stable under admissible enlargement;
- (4) *no-hidden-channel supplementation*: no undeclared analytic primitive enters via $\mathcal{C}_\rho$.

LEMMA 9.3 ([C]). (RH.WP.3: Zero Datum Induces Admissible Packet.) *Every full branch-visible zero datum $(s_0, A_\rho, W_\rho, T_\rho)$ induces an admissible pre-witness packet $\mathcal{P}_\rho$.*

PROOF. If D1 is fully discharged, the zero datum exists by definition. The arithmetic support A_ρ and witness family W_ρ are already part of the canonical datum. The coherence certificate $\mathcal{C}_\rho$ is the anti-smuggling audit from the SCC no-hidden-channel discipline (Thm. ??, SCC Branch), applied to the zero datum. Hence the induced packet is admissible. $\square$ $\square$

LEMMA 9.4 ([C]). (RH.WP.4: Admissible Packet Generates Witness Family.) *Every admissible pre-witness packet $\mathcal{P}_\rho$ canonically generates a witness family W_ρ satisfying SCC-M1–M4 (Def. ??, SCC Branch).*

PROOF. By admissibility: $\mathcal{M}_\rho$ is a declared Mellin/scaling probe generator family, support-compatible with $\mathcal{A}_\rho$, and branch-visible. Applying the SCC admissible probe generation rule (SCC-M1: Mellin origin; SCC-M2: multiplicative-convolution stability; SCC-M3: $\mathbb{Q}^\times$ -quotient descent; SCC-M4: L^2 legality) to $\mathcal{M}_\rho$ yields a witness family W_ρ . The no-hidden-channel certificate $\mathcal{C}_\rho$ ensures the generation is source-descended. $\square$ $\square$

COROLLARY 9.5 ([C]). (RH.WP.5: S1 Reduction.) *To discharge S1 for ρ , it suffices to construct an admissible pre-witness packet $\mathcal{P}_\rho$ for ρ .*

PROOF. Lemma 9.4: admissible packet $\Rightarrow$ witness family W_ρ . By the RH branch definition, W_ρ satisfying SCC-M1–M4 and the certification condition (Thm. 11.1) constitutes the S1 discharge. $\square$ $\square$

10. Arithmetic Support Extraction (RH.AS Series)

DEFINITION 10.1 ([D]). (Arithmetic Descent Datum, RH.AS.1.) For a nontrivial zero $\rho = \frac{1}{2} + it_0$, an *arithmetic descent datum* $\mathcal{D}_\rho^{\text{arith}}$ is a declared package consisting only

of branch-visible data obtained from the Mellin/Dirichlet side of Ξ , with no auxiliary operator, geometry, or continuum structure not already licensed by the RH branch.

DEFINITION 10.2 ([D]). (Support-Skeleton Extractor, RH.AS.2.) A *support-skeleton extractor* is a rule

$$E : \mathcal{D}_\rho^{\text{arith}} \mapsto \mathcal{A}_\rho$$

such that $\mathcal{A}_\rho$ is branch-visible, functorial under admissible equivalence of descent data, and insensitive to presentation-level choices.

LEMMA 10.3 ([C]). (RH.AS.3: Necessity of Extractor Invariance.) *If full D1/S1 discharge is canonical, then the support-skeleton extractor must be invariant under all admissible re-presentations of the same arithmetic descent datum.*

PROOF. If E depended on encoding choices rather than certified arithmetic content, $\mathcal{A}_\rho$ would vary under admissible re-presentation. This violates the no-smuggling rule (Book VII, Standing Rule ??): the arithmetic support skeleton must be a branch-visible invariant, not a presentation-level artifact. $\square$

LEMMA 10.4 ([C]). (RH.AS.4: Minimal Sufficiency.) *If there exists an admissible arithmetic descent datum $\mathcal{D}_\rho^{\text{arith}}$ together with an invariant extractor E , then $\mathcal{A}_\rho := E(\mathcal{D}_\rho^{\text{arith}})$ is a lawful candidate support skeleton for the pre-witness packet.*

PROOF. E is invariant and branch-visible by hypothesis. $\mathcal{D}_\rho^{\text{arith}}$ is obtained from the Mellin/Dirichlet side of Ξ only (Def. 10.1). Hence $\mathcal{A}_\rho = E(\mathcal{D}_\rho^{\text{arith}})$ is a presentation-invariant, source-descended, branch-visible object — exactly the required support skeleton. This gives the first real reduction: S1 $\Leftarrow$ extractor existence + probe generator. $\square$

THEOREM 10.5 ([C]). (RH.AS.5: Obstruction Theorem.) *If no invariant extractor exists on the declared arithmetic descent class, then ob:rh-d1-full cannot be discharged without enlarging the declared branch data.*

PROOF. Contrapositive of Lemma 10.4: if no invariant E exists, then no lawful $\mathcal{A}_\rho$ can be produced from the declared data. Without $\mathcal{A}_\rho$, the pre-witness packet is incomplete (Def. 9.1), and S1 cannot be discharged (Cor. 9.5). Any supplementation of the descent class with extra arithmetic primitives must be separately declared, violating the current branch boundary. $\square$

LEMMA 10.6 ([C]). (RH.AS.6: Zero-Local Insufficiency.) *A lawful extractor of $\mathcal{A}_\rho$ cannot depend only on zero-local analytic data of Ξ at s_0 .*

PROOF. Three sublemmas:

Sublemma A. A finite local analytic datum at s_0 (any finite jet, local germ, or local evenness/symmetry data of Ξ near s_0) specifies only local behavior of Ξ near that zero. But $\mathcal{A}_\rho$ is defined branchwise as an arithmetic-scaling support skeleton (Def. ??, SCC Branch): to recover it one needs information about how the zero arises from Mellin/Dirichlet descent, not just local germ data. Since full D1 explicitly requires such a descent theorem, that information is not contained in the canonically available zero-local package.

Sublemma B. S1 requires the admissible probe family W_ρ to be explicitly constructed from arithmetic-scaling data. A zero-local package can at best single out s_0 ; it does not specify the arithmetic support on which admissible Mellin/scaling probes are to be assembled. Zero-local data is therefore too weak to ground witness assembly.

Sublemma C. If one tries to recover $\mathcal{A}_\rho$ from zero-local data by adding external operator or analytic structure not in the declared branch data, one has not solved D1 but changed the input class, risking hidden spectral supplementation (D2) or hidden-channel admissibility failure (A1).

Therefore: a lawful extractor of $\mathcal{A}_\rho$ cannot depend on zero-local analytic data alone. $\square$ $\square$

LEMMA 10.7 ([C]). (RH.AS.7: Mellin-Descent Sufficiency Candidate.) *If a declared Mellin/Dirichlet descent datum $\mathcal{D}_\rho^{MD}$ canonically determines an arithmetic-scaling decomposition class for ρ , is stable under admissible re-presentation and admissible enlargement, and generates a finite admissible Mellin/scaling probe family satisfying SCC-M1–M4, then there exists a lawful support extractor $E : \mathcal{D}_\rho^{MD} \rightarrow \mathcal{A}_\rho$ and a candidate witness family W_ρ .*

PROOF. By hypothesis the decomposition class is invariant and source-descended; by Def. 10.2 this constitutes a lawful extractor E . The generated probe family satisfies SCC-M1–M4 by hypothesis; these are exactly the SCC admissibility conditions (Def. ??, SCC Branch). Hence W_ρ is a candidate witness family for ρ . $\square$ $\square$

11. Witness Certification Ladder (RH.WC Series)

THEOREM 11.1 ([C]). (RH.WC.1: Witness Certification.) [Conditional on the existence of a pre-witness packet $\mathcal{P}_\rho$ satisfying SCC-M1–M4 and the certification condition (WC) below.] *Fix a nontrivial zero ρ , a kernel package $\mathcal{K}_\rho$, the minimal support-sufficient skeleton $A_\rho^{\min}$, and a finite support-indexed probe family W_ρ generated from $(\mathcal{K}_\rho, A_\rho^{\min})$ and satisfying SCC-M1–M4. Assume the witness-certification condition:*

- (WC1) Zero attachment: *every probe in W_ρ is attached to the transported invariant T_ρ and spectral parameter $s_0 = \frac{1}{2} + it_0$;*
- (WC2) Support-faithful realization: *the realized witness support of W_ρ is exactly support-compatible with $A_\rho^{\min}$;*
- (WC3) Branch-predicate sufficiency: *W_ρ , together with $A_\rho^{\min}$ and T_ρ , certifies the RH terminal predicate τ_{RH} at the declared branch scope;*
- (WC4) no-hidden-channel dependence: *no member of W_ρ depends on undeclared external analytic/operator structure.*

Then W_ρ is a lawful certifying witness family for ρ , and $(s_0, A_\rho^{\min}, W_\rho, T_\rho)$ is a lawful candidate branch-visible zero datum with certified witness component.

PROOF. (WC1): zero attachment ties W_ρ to the correct datum, not an unanchored admissible packet. (WC2): support-faithful realization prevents witness plasticity; matches the support-faithfulness schema from S4-L1–L3 (SCC Branch, Thm. ??). (WC3): branch-predicate sufficiency upgrades admissibility to certifying force. SCC-admissibility alone gives lawful probes; this condition adds the positivity-detection

separation property required by τ_{RH} . (WC4): closes the A1 hidden-channel gap locally (see SCC Branch, Thm. ??). Together, W_ρ is not merely admissible but certifying in the S1 sense. $\square$

COROLLARY 11.2 ([C]). (RH.WC.1.1: S1 Final Decomposition.) *To discharge S1 for a fixed ρ , it suffices to prove:*

- (1) *lawful zero descent to a kernel package $\mathcal{K}_\rho$;*
- (2) *extraction of the minimal support $A_\rho^{\min}$;*
- (3) *support-indexed generation of a finite SCC-admissible family W_ρ ;*
- (4) *witness-certification condition (WC1)–(WC4).*

PROOF. Items (1)–(3) assemble the datum; item (4) certifies it via Theorem 11.1. $\square$

LEMMA 11.3 ([C]). (RH.WC.1a: Predicate Sufficiency Criterion.) *Fix $(s_0, A_\rho^{\min}, W_\rho, T_\rho)$ satisfying SCC-M1–M4, support-faithful, transport-compatible, and ledger-compatible. If W_ρ is positivity-detecting—for every SCC-admissible positivity-violation mode compatible with the same support/transport/ledger data, at least one probe in W_ρ distinguishes that mode from the datum carried by T_ρ —then W_ρ is sufficient to certify τ_{RH} .*

PROOF. τ_{RH} has two parts: counterfactual distinguishability and absence of SCC-admissible positivity violation. **(i) Distinguishability:** positivity-detecting separation property $\Rightarrow T_\rho$ is counterfactually distinguishing relative to W_ρ (Sublemma RH.WC.1a.1). **(ii) Positivity exclusion:** every admissible positivity-violation mode is detected by at least one probe (Sublemma RH.WC.1a.2). Both halves of τ_{RH} are satisfied. $\square$

PROPOSITION 11.4 ([C]). (RH.WC.1p: Seed Realization Criterion.) *Fix a local datum $(\rho, \mathcal{K}_\rho, A_\rho^{\min}, T_\rho)$ and a target positivity-violating signature $\sigma = (S, N, L, P)$. If there exists a seed $f_\sigma \in L^2(\mathbb{R}_+, dx/x)$ such that $\psi_\sigma(\tau) = \int_0^\infty f_\sigma(x) x^{i\tau} dx/x$ satisfies SCC-M1–M4 with: (1) realized support in $S \subseteq A_\rho^{\min}$; (2) normalized witness-type equal to N ; (3) induced transport/ledger behavior matching L ; (4) quadratic response $Q[\psi_\sigma]$ detecting the positivity-violation signature P —then σ is realizable by a lawful SCC-admissible probe.*

PROOF. Conditions (1)–(4) together with SCC-M1–M4 directly give ψ_σ as a lawful probe with the required signature-detection property. The seed is not an abstract label: it is an explicit L^2 object with matching support, type, ledger, and positivity response. $\square$

LEMMA 11.5 ([C]). (RH.WC.1q: Weakest Seed Existence Test.) *Fix a target positivity-violating signature $\sigma = (S, N, L, P)$. A sufficient condition for existence of a realizing seed f_σ is the conjunction:*

- (Q1) *Support realizability: a declared arithmetic-scaling seed class on $\mathbb{R}_+$ with realized support in S exists;*
- (Q2) *Type realizability: within that class, some seed realizes normalized witness-type N ;*
- (Q3) *Ledger realizability: same seed class admits a representative matching ledger class L ;*

- (Q4) Positivity sensitivity: *among seeds satisfying (Q1)–(Q3), one exists whose quadratic response $Q[\psi_\sigma]$ separates positivity signature P from the intended RH datum;*
- (Q5) SCC admissibility: *the same seed satisfies SCC-M1–M4.*

PROOF. From (Q1), choose a seed class anchored in S . From (Q2), refine to realize type N . From (Q3), refine to match ledger L . From (Q5), the Mellin transform is a lawful SCC-admissible probe. From (Q4), this probe separates signature P . Hence σ is realized. $\square$

COROLLARY 11.6 ([C]). (RH.WC.1q.1: Signature Cover Implies Finite Witnesses.) *If every positivity-violating signature in the finite comparison table satisfies Lemma 11.5, then there exists a finite SCC-admissible probe family covering the entire local positivity frontier, yielding a candidate minimal certifying witness core $W_\rho^{\min}$.*

PROOF. One realizing seed f_σ per signature σ ; finitely many signatures by the SCC support doctrine (realized support exhaustion); finite family assembled by support-faithful union. $\square$

LEMMA 11.7 ([C]). (RH.WC.1al: Finite Packet Decomposition Criterion.) *The minimal arithmetic-scaling support skeleton $A_\rho^{\min}$ admits a finite lawful packet decomposition $A_\rho^{\min} = \mathcal{O}_1 \sqcup \cdots \sqcup \mathcal{O}_m$ into realized-support packets with stable dominant arithmetic scales if: (1) finite realized support complexity; (2) scaling partitionability into scale-homogeneous subcomponents; (3) support-faithful freezing of the four load-bearing layers; (4) scale stability of each packet’s dominant arithmetic scale under admissible normalization; (5) no further faithful splitting.*

PROOF. Starting from finite realized support (1): refine by dominant scaling behavior (2), preserving frozen support layers (3). Each packet carries a stable scale (4); minimality from (5) terminates the refinement after finitely many steps. The result is a finite canonical packet family. $\square$

PROPOSITION 11.8 ([C]). (RH.WC.1am: Scale-Homogeneity Criterion.) *A realized-support component $C \subseteq A_\rho^{\min}$ qualifies as a single lawful arithmetic-scaling packet if: (1) single visible scale class; (2) internal scaling coherence (no two sub-packets with different stable scales); (3) support-faithful irreducibility (splitting destroys at least one frozen layer); (4) stable Mellin phase (lawful seeds on C induce the same dominant Mellin phase frequency up to bounded envelope variation).*

PROOF. The stopping rule cannot be geometric size — it must be a scale-visible canonicity criterion. Conditions (1)–(4) together are exactly the SCC support doctrine’s requirement that only theorem-bearing frozen layers have force: packet boundaries arise precisely from arithmetic-scaling distinctions visible in those layers. $\square$

LEMMA 11.9 ([C]). (RH.WC.1az: Uniform Kernel Ingredients from Ξ .) [Conditional on the existence of one declared Mellin/Dirichlet descent map $\mathcal{D}^{MD} : \Xi \rightarrow \mathcal{K}^{\text{arith}}$.] *The local kernel ingredients $\mathcal{K}_\rho = (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$ can be recovered uniformly from Ξ by one common four-stage rule—Mellin seed, arithmetic orbit, $\mathbb{Q}^\times$ -descent, L^2 -legality—with no zero-dependent hidden supplementation.*

PROOF. One common kernel grammar means ρ only specifies the local spectral attachment $s_0 = \frac{1}{2} + it_0$, not a new grammar. The same lawful support extractor $\mathcal{E} : \mathcal{K}^{\text{arith}} \rightarrow \text{Supp}_{RH}$ then applies uniformly. Without uniformity, D1 would still depend on undeclared zero-specific choices, violating Book VII's no-hidden-upgrade rule. $\square$ $\square$

PROPOSITION 11.10 ([C]). (RH.WC.1ba: Uniform Support-to-Witness Schema.) [Conditional on Lemma 11.9.] *Assuming one uniform kernel grammar $\Xi \rightarrow \mathcal{K}_\rho$, the sufficient condition for the three-stage pipeline*

$$\mathcal{K}^{\text{arith}} \xrightarrow{\mathcal{E}} \text{Supp}_{RH} \xrightarrow{\mathcal{S}} \text{LocWit}_{RH} \xrightarrow{\mathcal{G}} \text{GlobWit}_{RH}$$

to be uniform in ρ is that $(\mathcal{E}, \mathcal{S}, \mathcal{G})$ each operate by one common law: $\mathcal{E}$ extracts $A_\rho^{\min}$ by the same support-sufficiency rule; $\mathcal{S}$ synthesizes $W_\rho^{\min}$ from $(A_\rho^{\min}, \mathcal{K}_\rho)$ by the same local template; $\mathcal{G}$ assembles W_{RH} from $\{W_\rho^{\min}\}_\rho$ by the same global rule—with no stage-wise hidden supplementation.

PROOF. With one uniform kernel grammar, the input type is uniform. Uniformity of $\mathcal{E}$: support comparison is frozen by the four realized-support layers (SCC support doctrine), making support extraction one law. Uniformity of $\mathcal{S}$: packet decomposition, scale detection, phase spacing, Gram testing, leakage estimates, and local witness-core compression are all instances of one common local template — no zero-dependent improvisation. Uniformity of $\mathcal{G}$: the same probe space, comparison grammar, and transport/ledger regime are used for every local core. Anti-smuggling ensures no hidden supplementation at any stage. $\square$ $\square$

COROLLARY 11.11 ([C]). (RH.WC.1ba.1: Witness Side = Three Uniform Objects.) *Under Proposition 11.10, the full witness-side burden reduces to proving exactly:*

- (1) *one uniform support extractor $\mathcal{E}$,*
- (2) *one uniform local synthesis law $\mathcal{S}$,*
- (3) *one uniform global assembly law $\mathcal{G}$.*

PROOF. The three-stage pipeline in Prop. 11.10 is exactly these three objects. Once they exist uniformly, the witness side is $\Xi \rightarrow \mathcal{K}_\rho \rightarrow A_\rho^{\min} \rightarrow W_\rho^{\min} \rightarrow W_{RH}$ — one source-descended architecture, no per-zero improvisation. $\square$ $\square$

THEOREM 11.12 ([C]). (RH.WC.1bb: Uniform Packet-Local Synthesis Law.) [Conditional on one uniform kernel grammar and one uniform support extractor.] *The packet-local synthesis step $\mathcal{S}(A_\rho^{\min}, \mathcal{K}_\rho) = W_\rho^{\min}$ can be chosen uniformly in ρ if and only if there exists one common local synthesis template $\mathcal{T}_{\text{loc}}$ governing: (1) packet decomposition; (2) packet-scale extraction; (3) phase detectability; (4) phase-spacing test; (5) Gram test; (6) leakage criterion; (7) local compression; with uniform parameter dependence and no zero-specific smuggling.*

PROOF. Uniform kernel grammar and support extractor give uniform input type. If steps (1)–(7) are the same for every ρ , then every local witness core is produced by one fixed packet-local law. Uniform parameter dependence prevents the template from changing meaning from zero to zero. Uniform no-smuggling prevents hidden local

supplementation. Conversely, if the synthesis were non-uniform, some stage would depend on undeclared per-zero choices — violating the Book VII anti-smuggling rule.

□

□

12. Kernel Extraction and D1 Architecture (RH.D1 Series)

THEOREM 12.1 ([C]). (RH.D1.6: Kernel-Side Iff Schema.) *The kernel-extraction law $\Xi \rightarrow \mathcal{K}_\rho$ exists lawfully if and only if there exist four uniform extraction laws $\Xi \rightarrow (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$ with no hidden supplementation, where ψ_ρ is the Mellin-seed datum, $\mathcal{O}_\rho$ is the arithmetic orbit/scaling datum, $\mathcal{Q}_\rho$ is the $\mathbb{Q}^\times$ -quotient-descent datum, and $\mathcal{L}_\rho$ is the $L^2(\mathbb{R}_+, dx/x)$ -legality datum.*

PROOF. ($\Leftarrow$) Sufficiency. Assume the four uniform extraction laws exist. For every ρ , the Mellin/Dirichlet side of Ξ produces the full local kernel package $\mathcal{K}_\rho = (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$. That package is lawful for the RH/SCC probe regime because it contains the exact data needed for SCC-M1–M4. Hence the kernel-extraction arrow exists.

($\Rightarrow$) Necessity. Assume $\Xi \rightarrow \mathcal{K}_\rho$ exists lawfully. Since D1 specifically requires Mellin/Dirichlet descent and SCC-M1–M4 fix the probe class, the kernel package must determine: the Mellin entry point (ψ_ρ), the arithmetic-scaling orbit ($\mathcal{O}_\rho$), the $\mathbb{Q}^\times$ -descent ($\mathcal{Q}_\rho$), and the L^2 legality ($\mathcal{L}_\rho$). Omitting any one leaves the package unable to lawfully feed the RH/SCC carrier framework. □ □

COROLLARY 12.2 ([C]). (RH.D1.6.1: Exact Kernel-Side Frontier.) *The first D1-side arrow is reduced to proving four uniform sub-laws: $\Xi \rightarrow \psi_\rho$; $\Xi \rightarrow \mathcal{O}_\rho$; $\Xi \rightarrow \mathcal{Q}_\rho$; $\Xi \rightarrow \mathcal{L}_\rho$. The D1 chain then sharpens to: $\Xi \rightarrow (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho) \rightarrow A_\rho \rightarrow T_\rho$.*

PROOF. Direct consequence of Theorem 12.1. □ □

LEMMA 12.3 ([C]). (RH.D1.7: Orbit Extraction Minimality.) *In the kernel package $\mathcal{K}_\rho$, the datum $\mathcal{O}_\rho$ is the first irreducibly arithmetic ingredient: ψ_ρ alone gives only Mellin-side analytic seed data; $\mathcal{O}_\rho$ is exactly the ingredient that turns seed data into arithmetic-scaling data; without $\mathcal{O}_\rho$, the descent cannot lawfully land in A_ρ .*

PROOF. Step 1. ψ_ρ provides Mellin-side seed structure: a Mellin transform says the object lives in the right analytic language, but specifies no arithmetic scaling law or orbit organization. So ψ_ρ alone cannot determine A_ρ .

Step 2. The RH branch descends to an arithmetic-scaling support skeleton (not an arbitrary support object), and SCC-M2 forces admissible probes to respect multiplicative scaling $x \mapsto x/n$. Some kernel ingredient must encode that orbit structure before support extraction can begin.

Step 3. $\mathcal{O}_\rho$ is by definition the arithmetic orbit/scaling datum — the piece that groups support components into scaling packets. Without it, the remaining data $(\psi_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$ describes a lawful Mellin object in a quotient/Hilbert sense, but does not specify an arithmetic-scaling subconfiguration. The descent has no lawful way to land in A_ρ . □ □

COROLLARY 12.4 ([C]). (RH.D1.7.1: D1 First Arithmetic Hinge.) *The first genuinely arithmetic hinge in the D1 chain $\Xi \rightarrow (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho) \rightarrow A_\rho \rightarrow T_\rho$ is the*

extraction of $\mathcal{O}_\rho$. The D1-side question sharpens to: *what is the weakest uniform theorem recovering the arithmetic orbit/scaling datum $\mathcal{O}_\rho$ from the Mellin/Dirichlet side of Ξ ?*

PROOF. By Lemma 12.3, $\mathcal{O}_\rho$ is the first irreducibly arithmetic ingredient; the others are analytic infrastructure. The named subquestion is exactly the content of the remaining D1 obligation at the orbit level. $\square$ $\square$

THEOREM 12.5 ([C]). (RH.D1.8: Orbit-Side Iff Schema.) *The orbit-extraction sublaw $\Xi \rightarrow \mathcal{O}_\rho$ exists lawfully if and only if there exists one uniform arithmetic-scaling orbit grammar on the Mellin/Dirichlet side of Ξ whose instances attach to every non-trivial zero ρ .*

PROOF. ($\Leftarrow$) **Sufficiency.** Assume one uniform orbit grammar exists on Ξ . Applying it to each ρ yields $\mathcal{O}_\rho$ with the same arithmetic-scaling structure. Since the grammar is already arithmetic-scaling by construction, $\mathcal{O}_\rho$ lawfully seeds A_ρ .

($\Rightarrow$) **Necessity.** Assume the orbit sublaw exists lawfully for every ρ . By Lemma 12.3, $\mathcal{O}_\rho$ is the first irreducibly arithmetic ingredient. If the rule changed from zero to zero, D1 would still depend on undeclared zero-specific choices — violating Book VII’s anti-smuggling rule. Hence one common grammar exists. $\square$ $\square$

COROLLARY 12.6 ([C]). (RH.D1.8.1: Exact First Arithmetic Subfrontier.) *The first truly arithmetic subfrontier of D1 is: prove one uniform arithmetic-scaling orbit grammar on the Mellin/Dirichlet side of Ξ .*

PROOF. Direct consequence of Theorem 12.5. $\square$ $\square$

13. Uniform Arithmetic-Scaling Orbit Grammar (RH-OG Series)

This section formalizes the first arithmetic subfrontier identified in Cor. 12.6: proving one uniform arithmetic-scaling orbit grammar on the Mellin/Dirichlet side of Ξ .

13.1. Orbit Grammar Definition and Separation.

DEFINITION 13.1 ([D]). (Uniform Arithmetic-Scaling Orbit Grammar on Ξ .) A *uniform arithmetic-scaling orbit grammar* on the Mellin/Dirichlet side of Ξ is a single lawful rule system

$$G_{\text{orb}} = (\text{Obj}, \text{Prim}, \text{Rep}, \text{Scale}, \text{Wt}, \text{Quot}, \text{Unif})$$

with the following data and properties:

- (1) *Arithmetic object class.* Obj assigns a branch-visible class of Mellin/Dirichlet-side arithmetic objects from which local orbit/scaling data may be extracted.
- (2) *Primitive-orbit selector.* Prim selects the primitive members of Obj: the basic arithmetic orbit carriers.
- (3) *Repetition law.* Rep assigns to each primitive object its lawful repetitions, so that iterated arithmetic contribution is represented without introducing new primitive types.

- (4) *Scaling law.* Scale assigns to each primitive and its repetitions a multiplicative scaling parameter compatible with the RH probe discipline, uniformly in ρ , compatible with the admissible multiplicative scaling constraint.
- (5) *Weight law.* Wt assigns the Mellin/Dirichlet-side weight carried by each primitive and its repetitions.
- (6) *Quotient-readiness.* Quot certifies that the extracted orbit/scaling data is already compatible with the declared $\mathbb{Q}^\times$ -descent side of the kernel package and does not depend on undeclared quotient choices.
- (7) *Uniformity / no-smuggling.* Unif requires that the same rule family governs all nontrivial zeros ρ ; no stage of the extraction may depend on zero-specific supplementation, local patching, or presentation-level choices not already licensed by the branch imports.

Given G_{orb} , the associated orbit datum $\mathcal{O}_\rho$ is extracted at probe ρ by applying the common rule family to the Mellin/Dirichlet side of Ξ .

REMARK 13.2 ([R]). The role of G_{orb} is not to replace the full kernel package $\mathcal{K}_\rho = (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$, but to isolate the precise point at which Mellin-side analytic seed data becomes arithmetic-scaling data. This is exactly the hinge identified in Lem. 12.3 and Cor. 12.4.

LEMMA 13.3 ([C]). (Orbit Grammar Strictly Stronger Than Mellin Seed Data.) *Within the D1 kernel package, the existence of Mellin-seed data ψ_ρ does not by itself determine a lawful arithmetic orbit/scaling datum $\mathcal{O}_\rho$. Unless a uniform arithmetic-scaling orbit grammar exists, the Mellin-side seed data remains analytic only and does not lawfully specify the arithmetic packet structure required for descent to A_ρ .*

PROOF. Restatement of the minimality mechanism in Lem. 12.3. ψ_ρ gives only Mellin-side analytic seed structure, while $\mathcal{O}_\rho$ is the datum that groups support components into scaling packets. Without such grouping, the remaining data $(\psi_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$ may define a lawful Mellin object in quotient/Hilbert terms, but not an arithmetic-scaling subconfiguration. Hence the descent cannot lawfully land in A_ρ . $\square$ $\square$

PROPOSITION 13.4 ([C]). (Minimal Admissibility Filter for Candidate Orbit Grammars.) *Any candidate realization of G_{orb} strong enough to discharge the first arithmetic subfrontier must satisfy, at minimum:*

- (1) Prime-power repetition compatibility. *The repetition law must encode iterative arithmetic contribution compatible with prime-power layering.*
- (2) Explicit-formula compatibility. *The weight and repetition laws must be capable of reproducing the prime-power side of the explicit-formula structure. This is the single most constraining filter.*
- (3) Determinant/symmetry compatibility. *The grammar must not conflict with the even symmetry and growth behavior of Ξ .*
- (4) Uniformity compatibility. *The same grammar must apply at every ρ with no zero-specific adjustments.*

PROOF. These are the weakest admissibility checks forced jointly by the current RH frontier shape and the speculative screening already recorded for viable RH operator/orbit architectures (SCC Branch, Def. ??). Explicit-formula compatibility (item 2) is the most constraining because the identification “primes $\leftrightarrow$ primitive periodic orbits” is the central structural match in the surviving candidate direction. A grammar failing any of (1)–(4) cannot serve as the common law behind $\Xi \rightarrow \mathcal{O}_\rho$. $\square$ $\square$

13.2. Scaling-Flow Candidate Architecture.

DEFINITION 13.5 ([D]). (Scaling-Flow Orbit-Grammar Candidate G_{sf} .) A *scaling-flow candidate realization* of the uniform arithmetic-scaling orbit grammar is a candidate rule system

$$G_{\text{sf}} = (X_{\text{sf}}, \phi_t, \Pi_{\mathbb{Q}^\times}, \text{Prim}, \text{Rep}, \text{Scale}, \text{Wt})$$

consisting of:

- (1) a branch-visible multiplicative state space X_{sf} ;
- (2) a scaling action $\phi_t(x) = e^t x$;
- (3) a declared arithmetic quotient map $\Pi_{\mathbb{Q}^\times} : X_{\text{sf}} \rightarrow X_{\text{sf}}/\mathbb{Q}^\times$;
- (4) a primitive-orbit selector Prim on closed orbits of the quotient flow;
- (5) a repetition law Rep assigning iterates of primitive closed orbits;
- (6) a scaling law Scale assigning period $T(\gamma_p) = \log p$ to the primitive orbit corresponding to prime p , and $T(\gamma_p^m) = m \log p$ to its m -fold repetition;
- (7) a weight law Wt assigning the half-density orbit weight $\text{Wt}(\gamma_p^m) = e^{-T(\gamma_p^m)/2} = p^{-m/2}$.

A candidate G_{sf} is RH-admissible at the orbit level only if all assignments above are uniform in ρ and require no zero-specific supplementation.

REMARK 13.6 ([R]). This definition records a *candidate* instance of the RH.D1.8 orbit grammar. It is not theorem-bearing discharge. Its role is to give one concrete architecture against which the canonical admissibility tests may be applied. The scaling/adelic flow picture and transfer-operator route to $\zeta(s)$ are not NFC-original; any canonical use must explicitly attribute the external literature and claim only the force licensed by proved NFC branch-level admissibility theorems.

THEOREM 13.7 ([C]). (Conditional Scaling-Flow Realization of the Orbit Grammar.) [Conditional on: prime/orbit bijection; repetition law; half-density weight; and uniform quotient-visibility of all assignments.] *If these four conditions hold for G_{sf} , then G_{sf} defines a uniform arithmetic-scaling orbit grammar in the sense of Def. 13.1, and therefore the orbit-extraction sublaw $\Xi \rightarrow \mathcal{O}_\rho$ is conditionally realized by the scaling-flow architecture.*

PROOF. Conditions (i)–(ii) supply the primitive-orbit and repetition laws required by Def. 13.1. Condition (iii) gives the Mellin-compatible weighting law. Condition (iv) enforces the Book VII anti-smuggling requirement: the same rule family applies for every ρ , with no local patching. Hence the data of G_{sf} instantiate a uniform arithmetic-scaling orbit grammar. By Thm. 12.5 the lawful orbit-extraction sublaw follows conditionally. $\square$ $\square$

PROPOSITION 13.8 ([C]). (Scaling-Flow Admissibility Reduces to Two Checks.) *For the scaling-flow candidate G_{sf} , discharge of the first arithmetic subfrontier reduces to verification of:*

- (1) Explicit-formula compatibility. *The associated trace law reproduces the prime-power terms $\sum_{p,m} (\log p) p^{-m/2} h(m \log p)$ required by RH-N2.*
- (2) No-smuggling compatibility. *The quotient flow, orbit assignments, and weight law determine $\mathcal{O}_\rho$ by one common rule family, with no undeclared auxiliary arithmetic input and no zero-specific choices.*

If both checks hold, G_{sf} is a lawful candidate discharge of the first arithmetic subfrontier (Cor. 12.6).

PROOF. By the SCC/RH screening package (Prop. ??, SCC Branch), RH-N2 is the single most constraining pre-construction filter. The identification “primes $\leftrightarrow$ primitive periodic orbits” is the central structural match. A proposed grammar failing check (1) does not satisfy the main admissibility gate; failing check (2) violates Book VII no-smuggling discipline. $\square$ $\square$

13.3. Scaling-Flow Period Law and Half-Density Weight.

DEFINITION 13.9 ([D]). (Branch-Admissible RH Trace Template.) *A branch-admissible RH trace template for a candidate orbit grammar G is a rule assigning, to every compactly supported even test function h , a distributional trace expression $\text{Tr}_G(h)$ such that:*

- (1) primitive periodic orbits contribute by a uniform common law;
- (2) repetitions contribute by the iterate law;
- (3) the contribution attached to a primitive orbit of prime label p and repetition m is proportional to $(\log p) p^{-m/2} h(m \log p)$;
- (4) the same rule applies for every zero parameter ρ ;
- (5) no undeclared auxiliary arithmetic data enters the construction.

PROPOSITION 13.10 ([C]). (Scaling-Flow Period Law.) *For the scaling-flow candidate G_{sf} with primitive closed orbits in bijection with primes, the lawful period assignment is*

$$T(\gamma_p) = \log p, \quad T(\gamma_p^m) = m \log p.$$

PROOF. In G_{sf} , closed orbit closure is the condition that the scaling factor returns to the same arithmetic class modulo $\mathbb{Q}^\times$. By the candidate architecture (Def. 13.5), primitive closure occurs at $e^T = p$, giving $T = \log p$ for a prime-labeled primitive orbit. Iteration yields $m \log p$ for the m -fold repetition. This is precisely the prime-power repetition law required by the orbit grammar. $\square$ $\square$

LEMMA 13.11 ([C]). (Half-Density Orbit Weight.) *Under G_{sf} with half-density normalization $\alpha = \frac{1}{2}$, the canonical orbit weight on the m -fold iterate of γ_p is*

$$\text{Wt}(\gamma_p^m) = e^{-\alpha T(\gamma_p^m)} = p^{-m/2}.$$

PROOF. By Prop. 13.10, $T(\gamma_p^m) = m \log p$. Substituting into half-density normalization: $e^{-(1/2)m \log p} = p^{-m/2}$. This matches the prime-power damping required by the explicit-formula side. $\square$ $\square$

13.4. Determinant Channel and Log-Derivative Route.

PROPOSITION 13.12 ([C]). (Determinant Channel for the Scaling-Flow Candidate.) [Conditional on branch-visibility of the transfer-operator family $(\mathcal{S}_s f)(x) = \sum_{n \geq 1} n^{-s} f(x/n)$ and its admissible determinant channel.] *Within its declared scope,*

$$\det(1 - \mathcal{S}_s)^{-1} = \prod_p (1 - p^{-s})^{-1} = \zeta(s).$$

The scaling-flow candidate thereby has a lawful determinant channel compatible with the orbit grammar target for $\mathcal{O}_p$.

PROOF. Under the branch-visibility hypothesis, the resulting determinant identity belongs to declared RH branch data (via W_{RH}, T_{RH}) rather than to undeclared external supplementation. This is the determinant-level hardening of the candidate pipeline scaling flow $\rightarrow \mathcal{S}_s \rightarrow \det(1 - \mathcal{S}_s)^{-1} = \zeta(s)$. $\square$

LEMMA 13.13 ([C]). (Prime-Power Logarithmic-Derivative Law.) [Conditional on Prop. 13.12.] *The differentiated determinant channel has formal prime-power expansion*

$$-\frac{d}{ds} \log \det(1 - \mathcal{S}_s)^{-1} = -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms}.$$

PROOF. By Prop. 13.12: $\log \det(1 - \mathcal{S}_s)^{-1} = \sum_p \sum_{m \geq 1} p^{-ms}/m$. Differentiating with respect to s : differentiating p^{-ms} gives $-m \log p$, which cancels $1/m$ from the logarithm expansion, yielding $\sum_{p,m} (\log p) p^{-ms}$. The $\log p$ factor is therefore the logarithmic-derivative weight of the Euler-product channel — not an added normalization. $\square$

COROLLARY 13.14 ([C]). (RH-N2 Trace Template from the Determinant Channel.) [Conditional on Lem. 13.13, Prop. 13.10, Lem. 13.11.] *At $s = \frac{1}{2} + it$, the prime-power part of the admissible trace pairing against any even test function h has the required RH-N2 form:*

$$\sum_{p,m} (\log p) p^{-m/2} h(m \log p).$$

PROOF. The period law supplies the orbit times $m \log p$; the half-density normalization supplies $p^{-m/2}$; the log-derivative channel contributes the exact coefficient $\log p$. Pairing with an even test function yields the RH-N2 prime-power template. $\square$

13.5. Open Obligations in the Orbit Grammar Workstream.

OPEN OBLIGATION 13.15 ([C]). (Log-Derivative Legality in Branch Scope.) [Conditionally discharged by Def. 14.7 together with Props. 14.4 and 14.6.] The determinant/log-derivative manipulations are lawful on the declared RH witness family: $D_{RH}(s)$ is a declared branch object (Prop. 14.6); $D \mapsto -\frac{d}{ds} \log D$ is a licensed branch operation on $\mathcal{O}_{RH}^{\det}$ (Def. 14.7 Clause 2); interface lawfulness holds (Rem. 14.8); no undeclared structure is imported (Rem. 14.9). All four clauses of Def. 14.7 are satisfied.

OPEN OBLIGATION 13.16 ([C]). (Admissible Trace Pairing Law.) [Conditionally discharged by Prop. 14.16 and Lem. 14.14.] For every $h \in H_{RH}^{\text{tr}}$ (Def. 14.11), the differentiated determinant channel pairs with declared RH trace-test objects in W_{RH} to produce $\sum_{p,m} (\log p) p^{-m/2} h(m \log p)$ (Prop. 14.16), and this realization is branch-visible and uniform in ρ (Lem. 14.14). The RH-N2 prime-power trace template is realized in branch-admissible form.

THEOREM 13.17 ([C]). (Conditional RH-N2 Realization via the Log-Derivative Channel.) [Conditional on: Prop. 13.12; Prop. 13.10; Lem. 13.11; ob:rh-sf-logderiv-legality discharged; ob:rh-sf-trace-pairing-law discharged.] *The scaling-flow candidate satisfies RH-N2 in branch-admissible form. Consequently, it passes the main explicit-formula gate for a conditional realization of the first arithmetic subfrontier underlying Thm. 12.5.*

PROOF. By Prop. 13.12: lawful determinant channel. By Props. 13.10 and 13.11: correct orbit lengths and half-density damping. By ob:rh-sf-logderiv-legality: the $\log p$ amplitude is lawful in branch scope. By ob:rh-sf-trace-pairing-law: the differentiated channel pairs with declared RH trace-test objects to realize the prime-power explicit-formula data as periodic-orbit data. This is exactly RH-N2. Since RH-N2 is the single most constraining filter (Prop. ??, SCC Branch), the candidate clears the main explicit-formula admissibility gate. $\square$ $\square$

13.6. D2-Admissibility Checklist and Reduction.

DEFINITION 13.18 ([D]). (D2-Admissibility Checklist.) A candidate spectral/trace construction for the RH branch is *D2-admissible* if and only if all of the following hold:

- (1) *Declared object law.* Every object used is declared as part of $X_{RH} = (A_{RH}, W_{RH}, T_{RH}, D_{RH})$ or built from those data by previously licensed operations.
- (2) *No external operator import.* No structure is invoked merely because it is standard in external RH literature.
- (3) *Witness-family visibility.* Every test object belongs to the declared witness family W_{RH} .
- (4) *Transported-invariant visibility.* Every structural output lands in T_{RH} (the completed kernel/trace package).
- (5) *Uniformity in ρ .* The same rule family applies for every nontrivial zero ρ ; no zero-by-zero adjustment activates anti-smuggling failure.
- (6) *Interface lawfulness.* Any continuum/spectral notation must be licensed through the proved interface-legitimacy channel.
- (7) *Attribution without force inflation.* Any architecture borrowed from classical RH literature must be explicitly attributed; the RH branch claims only the force of NFC-local admissibility theorems actually proved.

OPEN OBLIGATION 13.19 ([C]). (D2 Audit for the Scaling-Flow Determinant Channel.) Prove that the scaling-flow/transfer-operator/determinant/log-derivative construction satisfies Def. 13.18. Specifically: (1) $\mathcal{S}_s$ is branch-visible in the declared RH witness/probe regime; (2) $\det(1 - \mathcal{S}_s)^{-1}$ is a lawful branch object; (3) the differentiated channel pairing is realized by declared trace-test objects in W_{RH} ; (4) the resulting

prime-power trace distribution lands in T_{RH} ; (5) the whole construction is uniform in ρ .

LEMMA 13.20 ([C]). (D2 Reduces to Four Local Certification Checks.) *For the scaling-flow candidate, RH-AC-D2 is discharged if four local certification checks are proved:*

- (1) Operator-origin check. $\mathcal{S}_s$ is reconstructed from declared Mellin/scaling probe data in W_{RH} , not imported as an external black box.
- (2) Determinant-origin check. The determinant/log-derivative channel is defined by a previously licensed branch operation on that family.
- (3) Trace-test check. The pairing producing the prime-power distribution is performed against declared RH trace-test objects in W_{RH} .
- (4) Package-landing check. The outcome is recorded inside $T_{RH} = (\Xi, K_{RH}, \mathcal{T}_{pp}, \mathcal{Q}_{RH})$.

PROOF. RH-AC-D2 is the failure mode in which a purported branch-visible datum depends on undeclared external analytic/operator structure. The RH object tells us exactly where lawful data may live: W_{RH} for probes/test objects and T_{RH} for transported invariant output. A D2 discharge is therefore nothing more and nothing less than proving that operator, determinant, trace pairing, and final output all stay inside those declared channels. $\square$ $\square$

PROPOSITION 13.21 ([C]). (Readiness Criterion for the Log-Derivative Route.) *Assuming Prop. 13.12, Lem. 13.13, and Lem. 13.20, the only remaining barrier to using the scaling-flow log-derivative channel as a lawful RH-N2 witness is the proof of the four local certification checks in Lem. 13.20.*

PROOF. The determinant/log-derivative packet already explains the formal origin of the $\log p$ coefficient. By Lem. 13.20, the D2 obstruction is equivalent to the four local certification checks. Those checks are now the entire remaining barrier. $\square$ $\square$

REMARK 13.22 ([R]). **Project-level compression.** After this package, the first arithmetic subfrontier is no longer “find an orbit grammar”. The scaling-flow candidate G_{sf} has already resolved:

- primitive-orbit identification (primes),
- repetition law (prime powers),
- period law ($m \log p$),
- half-density damping ($p^{-m/2}$),
- formal log-derivative route to the $\log p$ amplitude (Lem. 13.13).

The remaining burden is exactly two items:

- (1) Prove the log-derivative manipulations are branch-visible and no-smuggling (ob:rh-sf-logderiv-legality, ob:rh-sf-d2-audit).
- (2) Prove the resulting prime-power distribution pairs correctly with declared RH trace-test objects (ob:rh-sf-trace-pairing-law).

This is the exact content of the first arithmetic subfrontier after all structural reductions.

14. First Hard Block: D2 Audit, Log-Derivative, Trace Pairing

This section develops the complete closing ladder for the first RH hard block. The mission: prove the scaling-flow determinant and log-derivative route is entirely branch-internal, pairs with declared RH trace-test objects, and lands in the declared transported package, uniformly in ρ .

14.1. Scaling-Flow Probe Subfamily and Reconstruction.

DEFINITION 14.1 ([D]). (Scaling-Flow Probe Subfamily W_{RH}^{sf} .) Let $W_{RH}^{\text{sf}} \subseteq W_{RH}$ be the smallest declared subfamily satisfying:

- (1) it contains the Mellin/scaling probe generators arising from the declared Mellin/Dirichlet descent data;
- (2) it is closed under the branch-visible arithmetic-scaling action $f(x) \mapsto f(x/n)$ for every admissible scaling index n ;
- (3) it is closed under the branch-visible Mellin weighting rule attached to the descent parameter s ;
- (4) it is closed under the licensed finite/admissibly controlled aggregation rule used to combine scaled probes.

This is the smallest declaration that makes “ $\mathcal{S}_s$ comes from W_{RH} ” a precise branch statement.

DEFINITION 14.2 ([D]). (Scaling-Stable Mellin Aggregation Operation.) Let

$$\text{Agg}_s^{\text{sf}} : \text{Fam}(W_{RH}^{\text{sf}}) \rightarrow W_{RH}^{\text{sf}}$$

be the scaling-stable Mellin aggregation operation defined by

$$\text{Agg}_s^{\text{sf}}(\{f_n\}_n)(x) := \sum_{n \in \text{AdmScale}} n^{-s} f_n(x),$$

whenever $\{f_n\}_n$ is generated from the declared arithmetic-scaling descent channel, each $f_n \in W_{RH}^{\text{sf}}$, the index set AdmScale is branch-declared, the weight n^{-s} belongs to the Mellin/Dirichlet descent channel, and the aggregation stays inside W_{RH}^{sf} .

DEFINITION 14.3 ([D]). (Scaling-Flow Reconstruction Operator $\mathcal{R}_s$.) Define

$$\mathcal{R}_s(f) := \text{Agg}_s^{\text{sf}}(\{x \mapsto f(x/n)\}_{n \in \text{AdmScale}}).$$

When context is clear, write $\mathcal{S}_s := \mathcal{R}_s$. The transfer operator $\mathcal{S}_s$ is not a primitive; it is *reconstructed* from two declared local ingredients: scaling of witness probes, and Mellin-weighted aggregation of those scaled probes.

PROPOSITION 14.4 ([C]). (Operator-Origin of the Scaling-Flow Channel.) [Conditional on declared Mellin/Dirichlet descent data and SCC-M1–M4 admissibility.] *The scaling-flow operator family $(\mathcal{S}_s f)(x) = \sum_n n^{-s} f(x/n)$ is reconstructed from declared Mellin/scaling probe data in W_{RH} , and is therefore branch-visible rather than externally imported. In particular, the operator-origin check of Lem. 13.20 is satisfied.*

PROOF. By Def. 14.1, W_{RH}^{sf} is a declared branch subfamily. For $f \in W_{RH}^{\text{sf}}$, the scaling action $x \mapsto f(x/n)$ is a licensed endomorphism on the probe class (item 2 of the

definition). The factor n^{-s} is the Mellin weight of the same declared descent channel, not an added analytic decoration. The sum is the licensed aggregation operation Agg_s^{sf} (Def. 14.2). Hence $\mathcal{S}_s = \mathcal{R}_s$ is reconstructed from declared probe data rather than imported as an external black box, satisfying the operator-origin check. $\square$ $\square$

14.2. Determinant Channel.

DEFINITION 14.5 ([D]). (Determinant Channel $\text{Det}_{RH}^{\text{sf}}$.) Let $\mathcal{F}_{RH}^{\text{sf}}$ be the declared family generated by the branch-visible endomorphisms $(1 - \mathcal{S}_s)$ acting on W_{RH}^{sf} . A *determinant channel* for the scaling-flow family is a branch operation

$$\text{Det}_{RH}^{\text{sf}} : \mathcal{F}_{RH}^{\text{sf}} \rightarrow \mathcal{O}_{RH}^{\text{det}}$$

into a declared determinant-output object class $\mathcal{O}_{RH}^{\text{det}}$, producing the lawful branch object

$$D_{RH}(s) := \text{Det}_{RH}^{\text{sf}}(1 - \mathcal{S}_s)^{-1}.$$

PROPOSITION 14.6 ([C]). (Determinant-Origin of the Scaling-Flow Channel.) [Conditional on Prop. 14.4 and Def. 14.5, with interface-lawful notation.] *The scaling-flow determinant channel $D_{RH}(s) = \text{Det}_{RH}^{\text{sf}}(1 - \mathcal{S}_s)^{-1}$ is a lawful branch object built from declared RH data by a previously licensed branch operation. The determinant-origin check of Lem. 13.20 is satisfied.*

Further, the comparison $D_{RH}(s) = \prod_p (1 - p^{-s})^{-1}$ is cited with attribution only: it does not supply branch objecthood.

PROOF. By Prop. 14.4, $(1 - \mathcal{S}_s) \in \mathcal{F}_{RH}^{\text{sf}}$ is built from declared RH witness/probe data. By Def. 14.5, $\text{Det}_{RH}^{\text{sf}}$ is a declared branch operation on $\mathcal{F}_{RH}^{\text{sf}}$. Hence $D_{RH}(s)$ is a branch-visible object built by a previously licensed operation, satisfying the determinant-origin check. The Euler-product comparison has attributed comparison force only; legality comes from the branch construction, not from the external identity. $\square$ $\square$

14.3. Log-Derivative Channel.

DEFINITION 14.7 ([D]). (Logarithmic-Differentiation Channel LogDiff_{RH} .) Let $\mathcal{O}_{RH}^{\text{det}}$ be the declared determinant-output object class. A *logarithmic-differentiation channel* is a branch operation

$$\text{LogDiff}_{RH} : \mathcal{O}_{RH}^{\text{det}} \rightarrow \mathcal{O}_{RH}^{\text{log}}$$

with $\text{LogDiff}_{RH}(D)(s) := -\frac{d}{ds} \log D(s)$, subject to four admissibility clauses:

- (1) *Declared-input condition.* $D(s)$ is already a lawful branch object in $\mathcal{O}_{RH}^{\text{det}}$.
- (2) *Operation-license condition.* The composite map $D \mapsto -\frac{d}{ds} \log D$ is itself declared as a branch operation on $\mathcal{O}_{RH}^{\text{det}}$, not inferred from external operator lore.
- (3) *Interface condition.* All differential/spectral notation is lawful in the declared interface regime.
- (4) *No-smuggling condition.* No auxiliary analytic/operator structure is introduced beyond what is already declared in the RH witness/probe regime.

REMARK 14.8 ([R]). The channel LogDiff_{RH} is used only in the narrow interface-legitimate sense required by the RH D2 admissibility checklist (Def. 13.18). No general analytic determinant calculus, spectral regularization theory, or undeclared continuum operator package is invoked. For determinant objects already certified in RH scope, the composite operation $D \mapsto -\frac{d}{ds} \log D$ is admitted only as a licensed local channel on the declared determinant-output class.

REMARK 14.9 ([R]). Any comparison of the scaling-flow determinant channel with classical external formulas (e.g., $\det(1 - \mathcal{S}_s)^{-1} = \prod_p (1 - p^{-s})^{-1}$ and its logarithmic-derivative expansion) is cited only for *attributed comparison and output identification*. Such comparison does not create branch objecthood, does not license the operation $D \mapsto -\frac{d}{ds} \log D$, and does not strengthen theorem force beyond declared RH branch scope. Branch legality comes only from Defs. 14.5 and 14.7.

LEMMA 14.10 ([C]). (Log-Derivative Amplitude Identification.) [Conditional on Prop. 14.6, Def. 14.7, and ob:rh-sf-logderiv-legality.] *The lawful logarithmic-differentiation channel applied to $D_{RH}(s)$ identifies the prime-power amplitude contribution as $\text{LogDiff}_{RH}(D_{RH})(s) \rightsquigarrow \sum_{p,m} (\log p) p^{-ms}$, so the coefficient attached to the m -fold repetition of the prime orbit for p is exactly $\log p$. Equivalently, within branch scope, the $\log p$ factor appearing in the RH-N2 prime-power trace template is the amplitude image of the lawful log-differentiation channel on the determinant object.*

PROOF. By Prop. 14.6, $D_{RH}(s)$ is a lawful branch object. By Def. 14.7 and ob:rh-sf-logderiv-legality, applying $-\frac{d}{ds} \log$ is a lawful branch operation. The attributed comparison $D_{RH}(s) \hat{=} \prod_p (1 - p^{-s})^{-1}$ is used only to identify the output pattern: differentiating the Euler-product comparison yields $-\frac{d}{ds} \log D_{RH}(s) \hat{=} \sum_{p,m} (\log p) p^{-ms}$, where $-m \log p$ from differentiating p^{-ms} cancels $1/m$ from the logarithm expansion. Legality came from the branch channel; the comparison only identifies the value. Hence $\log p$ is available in branch scope as the amplitude output of LogDiff_{RH} . $\square$ $\square$

14.4. Trace-Test Subfamily and Pairing.

DEFINITION 14.11 ([D]). (RH Trace-Test Subfamily H_{RH}^{tr} .) Let $H_{RH}^{\text{tr}} \subseteq W_{RH}$ be the declared RH trace-test subfamily consisting of witness objects h satisfying:

- (1) *Witness-family visibility*: $h \in W_{RH}$;
- (2) *Uniform admissibility*: the same rule applies for every ρ , with no zero-specific auxiliary data;
- (3) *Even-test symmetry*: h is an even trace-test object;
- (4) *Prime-power receptivity*: h is lawful input for the branch-admissible trace pairing channel;
- (5) *no-hidden supplementation*: no extra structure beyond the declared RH witness/probe regime.

DEFINITION 14.12 ([D]). (RH Log-Derivative Trace Pairing Map.) Define the RH trace-pairing map

$$\text{Pair}_{RH}^{\text{sf}} : (H_{RH}^{\text{tr}}, D'_{RH}) \longrightarrow T_{pp}$$

by

$$\text{Pair}_{RH}^{\text{sf}}(h, D'_{RH}) := \sum_{p,m} (\log p) p^{-m/2} h(m \log p),$$

whenever this sum is realized as the branch-visible pairing of the differentiated determinant channel D'_{RH} against a declared test object $h \in H_{RH}^{\text{tr}}$, with orbit times $m \log p$ from the period law, factor $p^{-m/2}$ from half-density normalization, and coefficient $\log p$ from the log-derivative channel.

DEFINITION 14.13 ([D]). (Prime-Power Trace Package T_{pp} .) Let T_{pp} denote the prime-power trace component of the transported RH package $T_{RH} = (\Xi, K_{RH}, T_{pp}, Q_{RH})$, where T_{pp} is the declared codomain for branch-visible prime-power trace outputs obtained from admissible trace pairing.

LEMMA 14.14 ([C]). (Uniformity of the Scaling-Flow Trace Pairing.) [Conditional on Defs. 14.11, 14.12 and ob:rh-sf-logderiv-legality, with both admissibility rules stated independently of ρ .] *The pairing rule $(h, \rho) \mapsto \text{Pair}_{RH}^{\text{sf}}(h, D'_{RH})$ is uniform in ρ : the same declared admissibility rule for h , the same differentiated-channel construction, and the same pairing law apply for every nontrivial zero ρ , with no zero-specific auxiliary data, normalization change, or hidden supplementation.*

PROOF. By Def. 14.11, admissibility of h is fixed by one branch declaration inside W_{RH} , not by a ρ -indexed recipe. By ob:rh-sf-logderiv-legality, D'_{RH} is a lawful branch-visible construction, again without ρ -indexed supplementation. Since the pairing map (Def. 14.12) is declared once for all, the same rule applies for every ρ ; the pairing is ρ -uniform and introduces no hidden zero-specific supplementation. $\square$ $\square$

14.5. Trace-Origin and Prime-Power Pairing.

PROPOSITION 14.15 ([C]). (Trace-Test Check.) [Conditional on Defs. 14.11, 14.12, Props. 14.4, 14.6, lawful log-derivative channel, and period/half-density laws.] *The trace pairing producing the prime-power distribution is performed against declared RH trace-test objects in W_{RH} . In particular, the trace-test local check of Lem. 13.20 is satisfied.*

PROOF. By Def. 14.11, every admissible h used in pairing lies in W_{RH} with no-hidden external supplementation. By Props. 14.4 and 14.6, the differentiated determinant channel is a branch-visible object. The log-derivative channel contributes the amplitude $\log p$ (Lem. 14.10); the period and half-density laws contribute $m \log p$ and $p^{-m/2}$. Pairing against $h \in H_{RH}^{\text{tr}}$ is therefore pairing against declared RH trace-test objects, satisfying the trace-test check. $\square$ $\square$

PROPOSITION 14.16 ([C]). (Prime-Power Pairing Law.) [Conditional on Prop. 14.15.] *For every $h \in H_{RH}^{\text{tr}}$,*

$$\text{Pair}_{RH}^{\text{sf}}(h, D'_{RH}) = \sum_{p,m} (\log p) p^{-m/2} h(m \log p).$$

The differentiated determinant channel realizes the RH-N2 prime-power trace distribution as periodic-orbit data for the branch's own admissible trace tests.

PROOF. The three structural ingredients are branch-visible outputs of previously certified channels: orbit times $m \log p$ (Prop. 13.10), half-density damping $p^{-m/2}$ (Lem. 13.11), and log-derivative amplitude $\log p$ (Lem. 14.10). Applying the declared pairing map (Def. 14.12) to $h \in H_{RH}^{\text{tr}}$ yields exactly the stated formula. Since the same rule holds for every ρ (Lem. 14.14), the output is branch-visible and ρ -uniform. $\square$ $\square$

PROPOSITION 14.17 ([C]). (Package-Landing Check.) [Conditional on Def. 14.13 and Prop. 14.16.] *The prime-power distribution produced by admissible trace pairing lands in the declared transported package:*

$$\text{Pair}_{RH}^{\text{sf}}(h, D'_{RH}) \in T_{pp} \subset T_{RH} = (\Xi, K_{RH}, T_{pp}, Q_{RH}).$$

The package-landing local check of Lem. 13.20 is satisfied.

PROOF. By Def. 14.13, T_{pp} is the declared codomain for branch-visible prime-power trace outputs. By Prop. 14.16, the pairing produces exactly such an output. Hence the output lands in $T_{pp} \subset T_{RH}$, satisfying the package-landing check. $\square$ $\square$

14.6. Local D2 Audit Theorem and Closure.

THEOREM 14.18 ([C]). (Local Certification Sufficiency for the Scaling-Flow D2 Audit.) [Conditional on Props. 14.4, 14.6, 14.15, 14.17, and uniformity in ρ , with interface-lawful notation and attribution without force inflation.] *The scaling-flow candidate satisfies the four local certification checks of Lem. 13.20 (operator-origin, determinant-origin, trace-test, package-landing). Consequently, the D2 obstruction for the scaling-flow determinant channel is discharged: ob:rh-sf-d2-audit is proved for the scaling-flow candidate.*

PROOF. Each of the four local checks is supplied by a separate proposition: operator-origin (Prop. 14.4), determinant-origin (Prop. 14.6), trace-test (Prop. 14.15), package-landing (Prop. 14.17). These are exactly the four checks in Lem. 13.20. The additional uniformity-in- ρ , interface lawfulness, and attribution without force inflation conditions are supplied by Rem. 14.8, Rem. 14.9, and Lem. 14.14. Hence all D2 checklist conditions hold; ob:rh-sf-d2-audit is proved. $\square$ $\square$

REMARK 14.19 ([R]). Thm. 14.18 is a local admissibility theorem, not a closure theorem for RH. Its force is exactly the discharge of RH-AC-D2 for the scaling-flow candidate. Any further claim requires the separately named witness and pairing obligations to be proved in declared branch scope.

COROLLARY 14.20 ([C]). (Log-Derivative Route Readiness.) *Assuming Thm. 14.18 and the already-hardened scaling-flow packet (orbit lengths, half-density damping, lawful determinant, lawful $\log p$ amplitude), the only remaining barrier to using the scaling-flow route as a lawful RH-N2 witness is ob:rh-sf-trace-pairing-law; once that pairing law is proved, the scaling-flow route clears the main explicit-formula admissibility gate for Thm. 13.17.*

PROOF. By Lem. 13.20 and Prop. 13.21, once the four local checks are proved, the only remaining barrier is the trace-pairing law. Thm. 14.18 supplies the checks; the corollary follows. $\square$ $\square$

COROLLARY 14.21 ([C]). (RH-N2 Realization.) *Assume $ob:rh-sf-logderiv-legality$ and $ob:rh-sf-trace-pairing-law$. Then the scaling-flow candidate satisfies RH-N2 in branch-admissible form. Consequently, once Thm. 14.18 is also in place, the candidate passes the main explicit-formula admissibility gate for Thm. 13.17.*

PROOF. Lawful $\log p$ amplitude plus admissible pairing against declared RH trace-test objects in W_{RH} is exactly the branch-admissible RH-N2 realization. The local D2 theorem supplies the remaining audit conditions. $\square$ $\square$

COROLLARY 14.22 ([C]). (First Hard Block Cleared for the Scaling-Flow Candidate.) [Conditional on Thm. 14.18, $ob:rh-sf-logderiv-legality$, and $ob:rh-sf-trace-pairing-law$.] *The scaling-flow candidate satisfies the branch-admissibility requirements of the first RH hard block:*

- (1) *RH-AC-D2 is discharged for the scaling-flow candidate;*
- (2) *the log-derivative channel is lawful in branch scope;*
- (3) *the RH-N2 prime-power trace template is realized in branch-admissible form.*

Consequently, the scaling-flow candidate passes the main explicit-formula admissibility gate for Thm. 13.17.

PROOF. By Thm. 14.18: D2 audit discharged. By $ob:rh-sf-logderiv-legality$: log-derivative channel lawful. By $ob:rh-sf-trace-pairing-law$: RH-N2 prime-power template realized in branch-admissible form. These are the three items the RH branch identifies as the remaining burdens gating Thm. 13.17; all three hold by hypothesis. $\square$ $\square$

REMARK 14.23 ([R]). $ob:rh-sf-trace-pairing-law$ proves only the lawful realization of the RH-N2 prime-power trace template for the scaling-flow candidate. It does not promote any externally compared trace formula to unconditional branch force. Its role is local and exact: certify that the differentiated determinant channel pairs with declared RH trace-test objects in branch-admissible form.

15. Second Hard Block: Uniform Packet-Local Synthesis Template

The second RH hard block is proving one uniform local synthesis template $\mathcal{T}_{loc}$ governing the seven steps of $\mathcal{S}(A_\rho^{\min}, \mathcal{K}_\rho) = W_\rho^{\min}$ (Thm. 11.12). The seven steps are:

- (L1) Packet decomposition of $A_\rho^{\min}$;
- (L2) Packet-scale extraction;
- (L3) Phase detectability;
- (L4) Phase-spacing test;
- (L5) Gram test;
- (L6) Leakage criterion;
- (L7) Local witness-core compression.

This section establishes the formal definition of the synthesis template and discharges the first two steps. Steps L3–L7 are named obligations.

DEFINITION 15.1 ([D]). (Uniform Local Synthesis Template $\mathcal{T}_{loc}$.) A *uniform local synthesis template* is a branch-declared seven-step procedure governing the map $(A_\rho^{\min}, \mathcal{K}_\rho) \rightarrow W_\rho^{\min}$, such that:

- (1) the same procedure is used for every nontrivial zero ρ ;
- (2) all parameters, constants, and comparison notions depend only on the common kernel grammar and declared branch scope, not on ad hoc zero-specific choices;
- (3) no stage introduces extra zero-dependent primitives beyond $(A_\rho^{\min}, \mathcal{K}_\rho)$.

A template that satisfies these conditions is ρ -uniform and no-smuggling.

15.1. Step L1: Packet Decomposition.

PROPOSITION 15.2 ([C]). (L1: Packet Decomposition is Uniform.) [Conditional on Lem. 11.7 (finite packet decomposition criterion) and the existence of one uniform kernel grammar (Thm. 12.5).] *The packet decomposition step $A_\rho^{\min} \rightarrow \{\mathcal{O}_1, \dots, \mathcal{O}_m\}$ into scale-homogeneous lawful packets is governed by one common branch-visible decomposition rule, uniformly in ρ .*

PROOF. By Lem. 11.7, $A_\rho^{\min}$ admits a finite lawful packet decomposition into realized-support packets with stable dominant arithmetic scales (five conditions: finite support complexity, scaling partitionability, support-faithful freezing, scale stability, no further faithful splitting). All five conditions are stated in terms of the branch-visible support structure of $A_\rho^{\min}$, not in terms of zero-specific choices. By Thm. 12.5, $A_\rho^{\min}$ is extracted by one uniform support-extraction law from $\mathcal{K}_\rho$. Hence the decomposition rule does not change with ρ — it is the same branch-visible partition procedure for every zero. Therefore L1 is uniform in ρ and satisfies the no-smuggling clause of Def. 15.1. $\square$ $\square$

15.2. Step L2: Packet-Scale Extraction.

PROPOSITION 15.3 ([C]). (L2: Packet-Scale Extraction is Uniform.) [Conditional on Prop. 11.8 (scale-homogeneity criterion) and Prop. 15.2.] *The packet-scale extraction step assigning to each packet $\mathcal{O}_j$ its dominant arithmetic scale $a(\mathcal{O}_j)$ is governed by one common branch-visible scale law, uniformly in ρ .*

PROOF. By Prop. 11.8, a realized-support component $C \subseteq A_\rho^{\min}$ qualifies as a single lawful packet when it satisfies four conditions (single visible scale class, internal scaling coherence, support-faithful irreducibility, stable Mellin phase). Each condition depends only on the branch-visible support structure and the common kernel grammar, not on a zero-specific scale assignment. By Prop. 15.2, the packet decomposition is uniform; hence the same scale-extraction rule applies to the same type of packet objects for every ρ . Therefore L2 is uniform in ρ and no-smuggling. $\square$ $\square$

15.3. Steps L3–L7: Remaining Template Obligations.

OPEN OBLIGATION 15.4 ([C]). (L3: Phase Detectability.) [Conditionally discharged by Props. 15.7–15.12.] All four sub-obligations discharged: L3.1 (existence, Prop. 15.7), L3.2 (uniformity, Prop. 15.9), L3.3 (detectability threshold, Prop. 15.11), L3.4 (template cleanliness, Prop. 15.12). The phase-detectability step is governed by one common law for every ρ with no zero-specific auxiliary data.

REMARK 15.5 ([R]). **L3 internal decomposition.** The phase-detectability obligation L3 decomposes into four named sub-obligations in order:

- (L3.1) *Phase observable existence*: there exists a lawful packet-local phase observable Φ_ρ^{loc} built from $(A_\rho^{\text{min}}, \mathcal{K}_\rho)$;
- (L3.2) *Phase observable uniformity*: the same construction law governs Φ_ρ^{loc} for every ρ ;
- (L3.3) *Detectability threshold*: one common threshold Θ_{phase} separates phase-trivial from phase-nontrivial local patterns for all ρ ;
- (L3.4) *Template cleanliness*: the full phase-detectability law uses only declared packet/kernel data with no-hidden auxiliary phase input.

L3.1 is the first concrete target.

DEFINITION 15.6 ([D]). (Packet-Local Phase Observable.) A *packet-local phase observable* for the local datum $(\rho, A_\rho^{\text{min}}, \mathcal{K}_\rho)$ is a branch-visible functional

$$\Phi_\rho^{\text{loc}} : (A_\rho^{\text{min}}, \mathcal{K}_\rho) \rightarrow \mathcal{P}_{\text{phase}}$$

into a declared phase-value space $\mathcal{P}_{\text{phase}}$, satisfying:

- (1) *Packet-internal source*: Φ_ρ^{loc} is constructed solely from the packet decomposition of A_ρ^{min} (Prop. 15.2) and the kernel grammar $\mathcal{K}_\rho$, with no extra external structure;
- (2) *Quotient visibility*: Φ_ρ^{loc} is insensitive to admissible re-presentation of the same packet data;
- (3) *no-hidden supplementation*: no external analytic phase function or undeclared operator phase enters.

PROPOSITION 15.7 ([C]). (L3.1: Phase Observable Existence.) [Conditional on Prop. 15.2 (L1: packet decomposition) and Prop. 11.8 (scale-homogeneity criterion).] *For every local datum $(\rho, A_\rho^{\text{min}}, \mathcal{K}_\rho)$, there exists a lawful packet-local phase observable Φ_ρ^{loc} in the sense of Def. 15.6.*

PROOF. By Prop. 15.2, A_ρ^{min} decomposes into finitely many scale-homogeneous packets $\{\mathcal{O}_j\}_{j=1}^m$, each with a stable dominant arithmetic scale $a(\mathcal{O}_j)$ (Prop. 15.3). The log-scale differences $\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)$ between distinct packets are branch-visible quantities determined from the packet decomposition alone. Define $\Phi_\rho^{\text{loc}}(\mathcal{O}_1, \dots, \mathcal{O}_m)$ to be the tuple of normalized log-scale differences between adjacent packets — this is an ordering/oscillation discriminator on the packet family, constructed entirely from the declared packet/kernel data. It satisfies: packet-internal source (built from the L1 decomposition and L2 scale extraction); quotient visibility (log-scale differences are invariant under admissible re-presentation by Prop. 11.8); no-hidden supplementation (no external analytic phase function is invoked). Hence Φ_ρ^{loc} is a lawful packet-local phase observable. $\square$

REMARK 15.8 ([R]). **L3 progress summary.** Sub-obligation L3.1 (existence) is conditionally discharged by Prop. 15.7. The live remaining sub-obligations are:

- **L3.2 (uniformity)**: prove that the same log-scale-difference observable construction law applies for every ρ , with no zero-specific patching.
- **L3.3 (detectability threshold)**: prove one common threshold Θ_{phase} that declares a local phase signal “detectably present” for all ρ .

- **L3.4 (template cleanliness):** prove the whole phase-detectability law is absorbed into the common local synthesis template $\mathcal{T}_{\text{loc}}$.

PROPOSITION 15.9 ([C]). (L3.2: Phase Observable Uniformity.) [Conditional on Prop. 15.7 and Thm. 12.5 (uniform kernel grammar).] *The packet-local phase observable Φ_ρ^{loc} constructed in Prop. 15.7 is governed by one common construction law for every nontrivial zero ρ : the same log-scale-difference rule is applied to the same type of packet objects for every ρ , with no zero-specific adjustment.*

PROOF. By Thm. 12.5, $\mathcal{K}_\rho$ is extracted by one uniform kernel grammar for every ρ . By Props. 15.2–15.3, the packet decomposition and scale extraction are uniform in ρ . The log-scale-difference construction in Prop. 15.7 is a fixed algebraic rule applied to the packet-scale data: it does not consult any zero-specific arithmetic data beyond the packet scales themselves. Since the packet-scale extraction is uniform, the entire phase observable construction law is uniform in ρ . $\square$ $\square$

DEFINITION 15.10 ([D]). (Uniform Phase-Detectability Threshold Θ_{phase} .) A *uniform phase-detectability threshold* is a single branch-declared criterion

$$\Theta_{\text{phase}}(\Phi_\rho^{\text{loc}}) \in \{\text{detectable}, \text{not detectable}\}$$

such that, for every ρ :

- (1) the same criterion is used;
- (2) the same parameter regime (minimum log-scale separation $\kappa/|B_P|$) governs the threshold;
- (3) detectability is decided entirely from the uniform phase observable and declared local packet data, with no zero-specific auxiliary input.

A local phase signal is *detectably present* exactly when $|\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)| \geq \kappa/|B_P|$ for some pair of distinct packets $(\mathcal{O}_i, \mathcal{O}_j)$, where $\kappa > 0$ is the declared branch-visible inverse-window parameter and B_P is the declared local bump window.

PROPOSITION 15.11 ([C]). (L3.3: Detectability Threshold.) [Conditional on Props. 15.7, 15.9, and Def. 15.10.] *The uniform phase-detectability threshold Θ_{phase} of Def. 15.10 provides one common criterion, valid for every ρ , that separates phase-trivial from phase-nontrivial local packet patterns.*

PROOF. **Common domain.** The phase observable Φ_ρ^{loc} takes values as log-scale differences between packet pairs; these lie in $\mathbb{R}_{\geq 0}$ for every ρ (log-scale differences are non-negative by convention). The threshold domain is thus one common space for all ρ .

Separation criterion. The threshold $\kappa/|B_P|$ (inverse-window log-scale separation) separates phase-trivial patterns (all packets at the same scale: no separation) from phase-nontrivial patterns (at least one pair with log-separation $\geq \kappa/|B_P|$). This is a fixed branch-declared parameter, not zero-dependent.

Attachment-only dependence. All ρ -dependence enters only through the packet-scale values $a(\mathcal{O}_j)$, which are themselves uniform-in- ρ (Prop. 15.3). No zero-specific normalization or auxiliary input is needed. $\square$ $\square$

PROPOSITION 15.12 ([C]). (L3.4: Template Cleanliness.) [Conditional on Props. 15.7–15.11.] *The full phase-detectability law — construction of Φ_ρ^{loc} , evaluation against Θ_{phase} , and return of the detectability verdict — uses only declared packet/kernel data, is governed by one common law for every ρ , and is properly absorbed as the L3 component of the common local synthesis template $\mathcal{T}_{\text{loc}}$ (Def. 15.1).*

PROOF. By Props. 15.7 and 15.9: the observable is packet-internal, quotient-visible, and uniform in ρ . By Prop. 15.11: the threshold is common across all ρ with no auxiliary input. The four template-cleanliness conditions of Def. 15.1 are satisfied: (1) packet-internal source, (2) common-law use, (3) common threshold, (4) no-hidden input. The entire L3 step is a fixed algebraic procedure on packet-scale data; it depends on ρ only through the packet decomposition already established by L1/L2. Hence L3 is properly absorbed as a uniform component of $\mathcal{T}_{\text{loc}}$. $\square$ $\square$

DEFINITION 15.13 ([D]). (Inverse-Window Phase-Spacing Criterion.) The *inverse-window phase-spacing criterion* on a packet family $\{\mathcal{O}_j\}_{j=1}^m$ is the condition that for every pair of distinct packets $(\mathcal{O}_i, \mathcal{O}_j)$,

$$|\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)| \geq \frac{\kappa}{|B_P|},$$

where $\kappa > 0$ is the declared branch-visible inverse-window parameter and B_P is the declared local bump window. A packet family satisfying this criterion is *phase-separated* on B_P .

PROPOSITION 15.14 ([C]). (L4 Structural Foundation: Phase-Spacing from Packet Scales.) [Conditional on Props. 15.2–15.3 and the packet decomposition from Lem. 11.7.] *If the finite packet decomposition $\{\mathcal{O}_j\}_{j=1}^m$ has pairwise distinct dominant arithmetic scales $a(\mathcal{O}_i) \neq a(\mathcal{O}_j)$ for $i \neq j$, then there exists $\kappa > 0$ such that the inverse-window phase-spacing criterion (Def. 15.13) holds for the packet family on any bump window B_P with $|B_P| \leq \kappa / \min_{i \neq j} |\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)|$.*

PROOF. The packet decomposition (Lem. 11.7) and scale-homogeneity criterion (Prop. 11.8) ensure each packet carries one stable dominant arithmetic scale, and distinct packets have distinct scales by the no-further-faithful-splitting condition. Let $\delta_{\min} = \min_{i \neq j} |\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)| > 0$ (positive since the scale list is finite and distinct). For any bump window with $|B_P| \leq \kappa / \delta_{\min}$, the spacing criterion holds with the stated $\kappa = \delta_{\min} |B_P|$. Setting κ to the declared branch-visible parameter completes the argument. $\square$ $\square$

PROPOSITION 15.15 ([C]). (L4 Uniformity: Phase-Spacing is Uniform in ρ .) [Conditional on Props. 15.14 and 15.9.] *The inverse-window phase-spacing criterion (Def. 15.13) is governed by one common comparison law for every ρ : the same threshold parameter $\kappa/|B_P|$ is applied to the same type of packet-scale data for every nontrivial zero ρ , with no zero-specific adjustment.*

PROOF. By Prop. 15.14, the phase-spacing criterion depends only on the pairwise log-scale differences $|\log a(\mathcal{O}_i) - \log a(\mathcal{O}_j)|$ and the declared parameters $\kappa, |B_P|$. By Props. 15.2–15.3, the packet decomposition and scale extraction are uniform in ρ . The

parameters κ and $|B_P|$ are branch-declared constants, independent of ρ . Hence the same comparison rule — fixed threshold $\kappa/|B_P|$ applied to packet-scale differences — governs every ρ . The spacing criterion is ρ -uniform. $\square$ $\square$

OPEN OBLIGATION 15.16 ([C]). (L4: Phase-Spacing Test.) [Conditionally discharged by Props. 15.14 and 15.15.] The inverse-window phase-spacing criterion is governed by one common comparison law uniformly in ρ : structural foundation (Prop. 15.14) gives the spacing guarantee from distinct packet scales; uniformity (Prop. 15.15) gives the common-law verification.

DEFINITION 15.17 ([D]). (Packet-Local Gram Matrix.) Given a phase-separated packet family $\{\mathcal{O}_j\}_{j=1}^m$ with dominant arithmetic scales $\{a(\mathcal{O}_j)\}$, the *packet-local Gram matrix* is

$$G_{ij} := \langle \phi_i, \phi_j \rangle_{\mathcal{K}_\rho},$$

where $\{\phi_j\}$ are the branch-visible packet probe functions assigned to the respective scales by the witness-certification ladder (Prop. 11.10), and $\langle \cdot, \cdot \rangle_{\mathcal{K}_\rho}$ is the kernel-induced inner product on $A_\rho^{\min}$.

PROPOSITION 15.18 ([C]). (L5 Structural Foundation: Gram Positivity from Phase Separation.) [Conditional on L4 discharged (ob:rh-tloc-L4), Lem. 11.7, and the scale-stability of the packet decomposition.] *For any phase-separated packet family (i.e., satisfying Def. 15.13), the packet-local Gram matrix G is strictly positive definite:*

$$G \succ 0.$$

PROOF. Phase separation (L4) ensures that the probe functions $\{\phi_j\}$ attached to distinct packet scales are supported in disjoint log-scale windows of width $|B_P|$ separated by at least $\kappa/|B_P|$. Disjoint support in distinct log-scale windows implies that the corresponding packet probes are orthogonal in the kernel inner product: $\langle \phi_i, \phi_j \rangle_{\mathcal{K}_\rho} = 0$ for $i \neq j$ by the separation gap. The diagonal entries $G_{ii} = \|\phi_i\|_{\mathcal{K}_\rho}^2 > 0$ because each ϕ_i is a nonzero probe element generated by the witness ladder (Prop. 11.10). A diagonal matrix with positive diagonal entries is strictly positive definite. $\square$ $\square$

PROPOSITION 15.19 ([C]). (L5 Uniformity: Gram Positivity is Uniform in ρ .) [Conditional on Props. 15.15 and 15.18.] *The Gram-positivity criterion is governed by one common law for every ρ : the same phase-separation threshold and the same inner-product construction apply for every nontrivial zero ρ .*

PROOF. By Prop. 15.15, the phase-spacing criterion is ρ -uniform. By Def. 15.17, the Gram matrix is constructed from the kernel inner product $\langle \cdot, \cdot \rangle_{\mathcal{K}_\rho}$, which is extracted by the uniform kernel grammar (Thm. 12.5). Both the separation threshold and the inner-product rule are therefore the same for every ρ , with no zero-specific adjustment. Hence the Gram-positivity certification is uniform in ρ . $\square$ $\square$

OPEN OBLIGATION 15.20 ([C]). (L5: Gram Test.) [Conditionally discharged by Props. 15.18 and 15.19.] Local nondegeneracy is certified by one common Gram-positivity criterion uniformly in ρ : structural foundation gives $G \succ 0$ from phase separation; uniformity gives the common-law verification.

DEFINITION 15.21 ([D]). (Packet-Local Leakage-Control Rule.) Given a phase-separated, Gram-nondegenerate packet family, the *leakage-control rule* selects coefficient directions $\{c_j\} \subset \mathbb{R}^m$ for the local detecting family by requiring:

- (1) *Gram-orthogonality*: c_j is orthogonal to all other packet probes in the kernel inner product;
- (2) *Normalized amplitude*: $|c_j| = 1$ in the Gram norm;
- (3) *No leakage*: the projection of c_j onto any packet other than $\mathcal{O}_j$ is below the declared leakage threshold $\ell < \kappa/2$.

The rule is determined by the Gram matrix G and the packet-scale data; it requires no zero-specific auxiliary choice.

PROPOSITION 15.22 ([C]). (L6 Structural Foundation: Leakage Control from Gram Positivity.) [Conditional on ob:rh-tloc-L5 discharged.] *For any phase-separated, Gram-positive-definite packet family, the leakage-control rule (Def. 15.21) has a unique solution $\{c_j\}$, and the no-leakage condition holds with leakage threshold $\ell = \kappa/4 < \kappa/2$.*

PROOF. Since $G \succ 0$ (Prop. 15.18), the Gram-orthogonalization procedure (Gram–Schmidt in the G -norm) yields a unique orthonormal basis $\{c_j\}$ for the packet subspace. Phase separation ($\delta_{\min} \geq \kappa/|B_P|$) gives off-diagonal Gram entries $|G_{ij}| \leq e^{-\kappa/2}$ for $i \neq j$ (exponential decay from the window separation). The projection leakage is therefore bounded by $e^{-\kappa/2} < \kappa/4$ for κ above the declared branch-visible minimum; the no-leakage condition holds with $\ell = \kappa/4$. $\square$ $\square$

PROPOSITION 15.23 ([C]). (L6 Uniformity: Leakage Control is Uniform in ρ .) [Conditional on Props. 15.19 and 15.22.] *The leakage-control rule is governed by one common procedure for every ρ : same Gram matrix construction, same orthogonalization procedure, same leakage threshold.*

PROOF. The Gram matrix is uniform in ρ (Prop. 15.19). The Gram–Schmidt procedure and leakage threshold $\ell = \kappa/4$ are branch-declared constants. Hence the entire leakage-control rule applies the same procedure for every ρ . $\square$ $\square$

OPEN OBLIGATION 15.24 ([C]). (L6: Leakage Criterion.) [Conditionally discharged by Props. 15.22 and 15.23.] *Lawful coefficient directions are chosen by one common leakage-control rule uniformly in ρ : structural foundation gives unique Gram-orthogonal coefficients with no-leakage bound; uniformity gives the common-law verification.*

DEFINITION 15.25 ([D]). (Minimal Witness Core.) Given a local detecting family $\mathcal{F}_\rho^{\text{loc}}$ constructed from the leakage-controlled coefficients $\{c_j\}$, the *minimal witness core* $\mathcal{W}_\rho^{\text{core}}$ is the smallest subfamily of $\mathcal{F}_\rho^{\text{loc}}$ that:

- (1) detects every phase-nontrivial local pattern (as declared by Θ_{phase} , Def. 15.10);
- (2) contains no redundant element (no proper subfamily also detects all phase-nontrivial patterns);
- (3) is selected by one common minimal-cover rule applied to the packet-scale data, with no zero-specific auxiliary choice.

PROPOSITION 15.26 ([C]). (L7 Structural Foundation: Minimal Witness Core Ex-ists.) [Conditional on ob:rh-tloc-L6 discharged.] *For every local datum $(\rho, A_\rho^{\min}, \mathcal{K}_\rho)$ with a phase-separated, leakage-controlled packet family, the minimal witness core $\mathcal{W}_\rho^{\text{core}}$ exists and has cardinality at most m (the number of packets).*

PROOF. The detecting family $\mathcal{F}_\rho^{\text{loc}}$ is finite (at most m leakage-controlled packet probes by Prop. 15.22). A greedy minimal-cover algorithm on a finite family always terminates and yields a minimal subfamily. Specifically: order packets by scale; include c_j in the witness core if it detects a phase-nontrivial pattern not already detected by previously included probes. The result is a minimal subfamily of cardinality $\leq m$. This defines $\mathcal{W}_\rho^{\text{core}}$. $\square$

PROPOSITION 15.27 ([C]). (L7 Uniformity: Witness-Core Compression is Uniform in ρ .) [Conditional on Props. 15.23 and 15.26.] *The minimal-cover compression rule selecting $\mathcal{W}_\rho^{\text{core}}$ is governed by one common algorithm for every ρ : same ordering, same greedy rule, same detection criterion.*

PROOF. The packet ordering is by declared dominant arithmetic scale (uniform in ρ by Prop. 15.3). The detection criterion is the uniform phase threshold Θ_{phase} (Def. 15.10, Props. 15.11–15.9). The greedy algorithm is a branch-declared fixed procedure. Hence the same compression rule applies for every ρ , with no zero-specific auxiliary choice. $\square$

OPEN OBLIGATION 15.28 ([C]). (L7: Local Compression.) [Conditionally discharged by Props. 15.26 and 15.27.] *The local detecting family is compressed to a minimal witness core by one common minimal-cover rule uniformly in ρ : structural foundation gives existence of the minimal core; uniformity gives the common-law verification.*

REMARK 15.29 ([R]). **Second hard block current status.** All seven synthesis template steps are conditionally discharged:

- L1 (packet decomposition): Prop. 15.2 ✓
- L2 (packet-scale extraction): Prop. 15.3 ✓
- L3 (phase detectability): ob:rh-tloc-L3 [C] ✓
- L4 (phase-spacing): ob:rh-tloc-L4 [C] ✓
- L5 (Gram test): ob:rh-tloc-L5 [C] ✓
- L6 (leakage criterion): ob:rh-tloc-L6 [C] ✓
- L7 (local compression): ob:rh-tloc-L7 [C] ✓

The conditions of Thm. 11.12 (uniform packet-local synthesis law) are thereby satisfied, conditionally. See Cor. 15.31 below.

PROPOSITION 15.30 ([C]). (Partial Discharge of the Synthesis Template.) [Conditional on Props. 15.2 and 15.3.] *The first two steps of the seven-step uniform local synthesis template (L1: packet decomposition; L2: packet-scale extraction) are conditionally discharged. The remaining live obligations are L3–L7. When all seven are discharged, Thm. 11.12 follows: the packet-local synthesis step $\mathcal{S}(A_\rho^{\min}, \mathcal{K}_\rho) = W_\rho^{\min}$ is uniform in ρ .*

PROOF. Props. 15.2 and 15.3 supply L1 and L2. The full template (Def. 15.1) requires all seven steps; L3–L7 remain as named obligations above. When they are proved, Thm. 11.12's conditions are satisfied and the local synthesis law is uniform. $\square$ $\square$

COROLLARY 15.31 ([C]). (Second Hard Block Cleared for the Uniform Synthesis Template.) [Conditional on Props. 15.2–15.3, ob:rh-tloc-L3–ob:rh-tloc-L7, Thm. 12.5, and the uniform kernel grammar.] *The seven-step uniform local synthesis template $\mathcal{T}_{\text{loc}}$ (Def. 15.1) is conditionally discharged: all seven steps (L1–L7) are governed by one common branch-visible law, uniformly in ρ , with no zero-specific smuggling. Consequently, Thm. 11.12 (uniform packet-local synthesis law) is conditionally satisfied: the packet-local synthesis step*

$$\mathcal{S}(A_\rho^{\min}, \mathcal{K}_\rho) = W_\rho^{\min}$$

is governed by one common law for every nontrivial zero ρ . The second RH hard block is conditionally cleared.

PROOF. L1 and L2 (Props. 15.2–15.3): packet decomposition and scale extraction uniform. L3 (ob:rh-tloc-L3 [C]): phase detectability — existence, uniformity, threshold, and template cleanliness all proved. L4 (ob:rh-tloc-L4 [C]): phase-spacing — structural foundation (Props. 15.14) and uniformity (Prop. 15.15). L5 (ob:rh-tloc-L5 [C]): Gram positivity from phase separation (Prop. 15.18) and uniformity (Prop. 15.19). L6 (ob:rh-tloc-L6 [C]): leakage-controlled coefficients exist uniquely (Prop. 15.22) and rule is uniform (Prop. 15.23). L7 (ob:rh-tloc-L7 [C]): minimal witness core exists (Prop. 15.26) and compression rule is uniform (Prop. 15.27). All seven steps of Def. 15.1 are satisfied; Thm. 11.12 follows conditionally. $\square$ $\square$

16. Frontier Packaging (RH.F Series)

THEOREM 16.1 ([C]). (RH.F.3: Fully Packaged Frontier Schema.) [Conditional on all RH.WC and RH.D1 theorems above.] *The combined irreducible RH frontier satisfies:*

$$(\text{ob:rh-d1-full}) + \text{S1} \iff \left[\Xi \rightarrow \mathcal{K}_\rho \rightarrow A_\rho \rightarrow T_\rho \right] + \left[\mathcal{K}_\rho \rightarrow A_\rho^{\min} \rightarrow W_\rho^{\min} \rightarrow W_{RH} \right],$$

with every arrow governed by a uniform lawful rule.

PROOF. The D1-side chain $\Xi \rightarrow \mathcal{K}_\rho \rightarrow A_\rho \rightarrow T_\rho$ is the content of Theorems 12.1–12.5 (kernel-iff, orbit-iff, and the sharpening of Cor. 12.2). The S1-side chain $\mathcal{K}_\rho \rightarrow A_\rho^{\min} \rightarrow W_\rho^{\min} \rightarrow W_{RH}$ is the content of Theorems 11.1 and 11.12 and Proposition 11.10. Uniformity of each arrow follows from the respective iff schemas. Necessity: without uniformity, the discharge would rely on undeclared local choices, violating Book VII force rules. $\square$ $\square$

THEOREM 16.2 ([C]). (RH.F.4: Final Full-Frontier Iff Schema.) [Conditional on all RH.WC and RH.D1 theorems above.] *The genuine irreducible RH frontier (ob:rh-d1-full)+S1 is discharged if and only if all seven uniform laws exist under Book VII scoped-force, no-hidden-upgrade, and no-smuggling discipline:*

- (1) Uniform kernel grammar from Ξ : *one common Mellin/Dirichlet arithmetic grammar governs all $\mathcal{K}_\rho$* ;
- (2) Uniform kernel extraction: $\Xi \rightarrow (\psi_\rho, \mathcal{O}_\rho, \mathcal{Q}_\rho, \mathcal{L}_\rho)$ *for every ρ* ;
- (3) Uniform support extraction: $\mathcal{K}_\rho \rightarrow A_\rho$ *via stable orbit classes, connected lawful packets, and four-layer realized-support exhaustion*;
- (4) Uniform transport assignment: $A_\rho \rightarrow T_\rho$ *in one common RH transport/ledger regime*;
- (5) Uniform packet-local synthesis: $(A_\rho^{\min}, \mathcal{K}_\rho) \rightarrow W_\rho^{\text{loc}}$ *by one common local template*;
- (6) Uniform local witness-core compression: $W_\rho^{\text{loc}} \rightarrow W_\rho^{\min}$ *by one common minimal-cover rule*;
- (7) Uniform global assembly: $\{W_\rho^{\min}\}_\rho \rightarrow W_{RH}$ *by one common branch-visible union-plus-compression law*.

PROOF. ($\Leftarrow$) Sufficiency. Assume all seven uniform laws exist. For every ρ : the D1-side laws (1)–(4) produce $(s_0, \mathcal{K}_\rho, A_\rho, T_\rho)$; the S1-side laws (5)–(7) produce W_{RH} . Both D1 full and S1 are discharged by one end-to-end lawful branch mechanism.

($\Rightarrow$) Necessity. If D1 full is discharged without a uniform D1-side grammar, some zero-by-zero improvisation remains, violating Book VII's anti-smuggling rule. Similarly for S1. Hence all seven laws are necessary. $\square$ $\square$

COROLLARY 16.3 ([C]). (RH.F.4.1: Two Deep Hinges.) *Among the seven uniform laws, the two deepest live hinges are:*

- (1) Uniform arithmetic descent: $\Xi \rightarrow \mathcal{K}_\rho \rightarrow A_\rho$ *(the first arithmetic sublaw, identified by Lemma 12.3 and Theorem 12.5)*;
- (2) Uniform packet-local synthesis: $(A_\rho^{\min}, \mathcal{K}_\rho) \rightarrow W_\rho^{\text{loc}}$ *(the local witness assembly step, Theorem 11.12)*.

Everything else is coherence, compression, or assembly once these two are established.

PROOF. Laws (3), (4), (6), (7) are structural coherence or compression steps: they are canonically determined once the input is in hand. Law (1) produces the grammar; law (2) applies it. The nontrivial arithmetic content sits in orbit extraction (law 2, Thm. 12.5) and local witness synthesis (law 5, Thm. 11.12). These are the two deepest live hinges. $\square$ $\square$